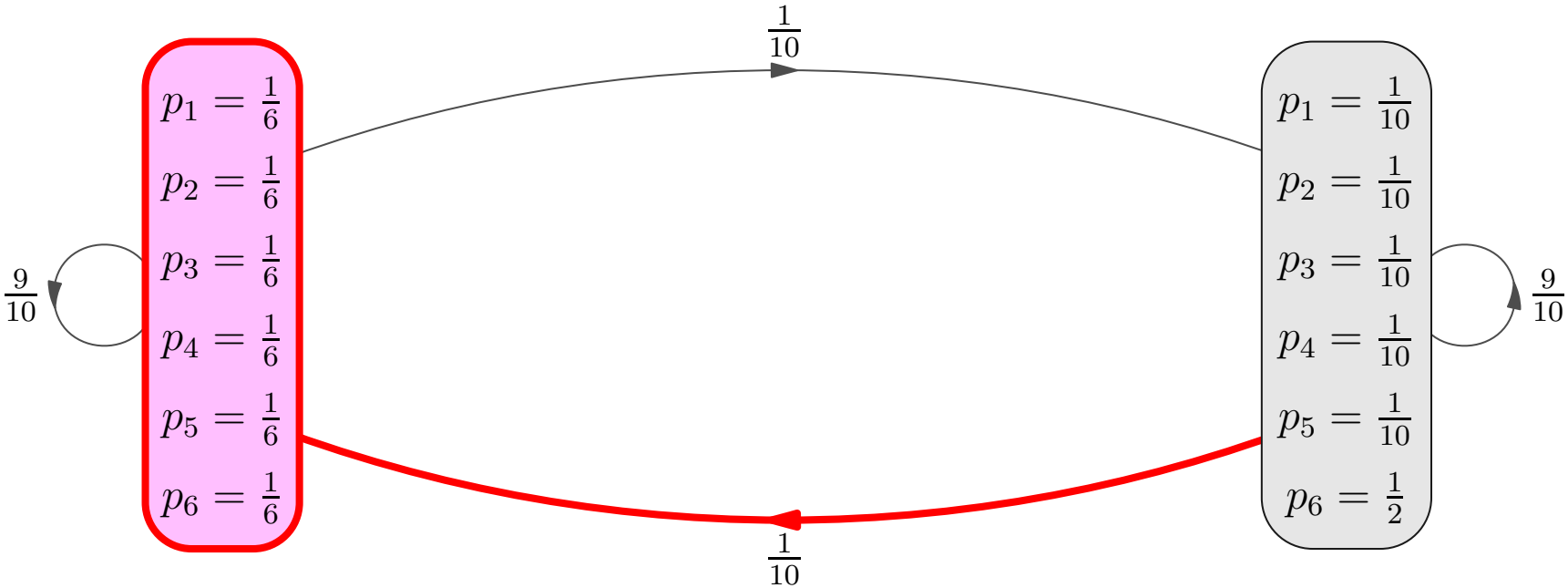
# Lecture 7: Hidden Markov Models

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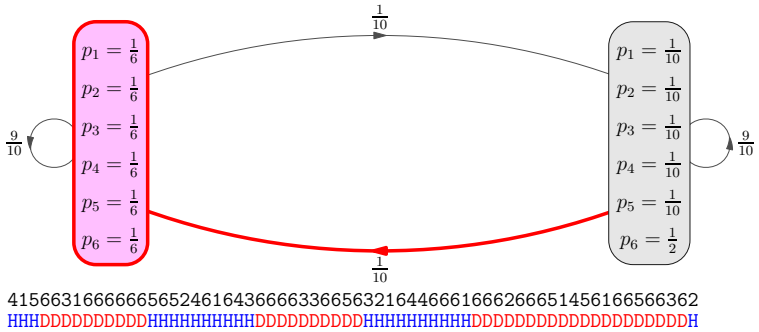
HHHDDDDDDDDDDHHHHHHHHHHHDDDDDDDDDDHHHHHHHHHHHDDDDDDDDDDDDDDDDDDDDH

*Generative Models, Markov Chains, FSA, HMMs*

# Outline

1. **Probabilistic Models**
2. Hidden Markov Models
3. Summing Over Paths



# Probabilistic Models

- Probability is the natural language for talking about uncertainty
- Reason about biological sequences where we have uncertainty
- Can we develop methods to answer questions such as how likely is a certain sequence to be a coding region?
- Or what is the probability that this part of a protein forms a beta sheet?

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# Probabilistic Models and ML

- In machine learning we usually have a data set consisting of pairs of features, $\boldsymbol{x}$, and targets, $y$
- Many machine learning problems comes down to determining the probability of the target $y$ given the features, $\boldsymbol{x}$

$$\mathbb{P}\left(y|\boldsymbol{x}\right)$$

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- However, we usually just use a model with lots of parameters, $\boldsymbol{\theta}$ and train on data to find $\mathbb{P}\left(y|\boldsymbol{x},\boldsymbol{\theta}\right)$

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# Generative Models

- If we really want to use a principled model then we need to construct probabilistic models by thinking about how the data is generated

- This involves computing the joint probability

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- We can still do discrimination since

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# Markov Models

- A lot of biological data involves sequences, e.g. DNA, proteins
- The simplest model for a sequence is a Markov model, where the probability of the next element in the sequence just depends on the current state

$$\mathbb{P}\left(x_t | x_{t-1},\, x_{t-2},\, \ldots,\, x_1\right) = \mathbb{P}\left(x_t | x_{t-1}\right)$$

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# Finite State Automaton

- As we saw in the last lecture, one mechanism to make more complex models is to construct a finite state automaton
- For sequences we can make the probability of generating a character from some alphabet depend on the state
- We can then build a model describing the probability of transitioning between states
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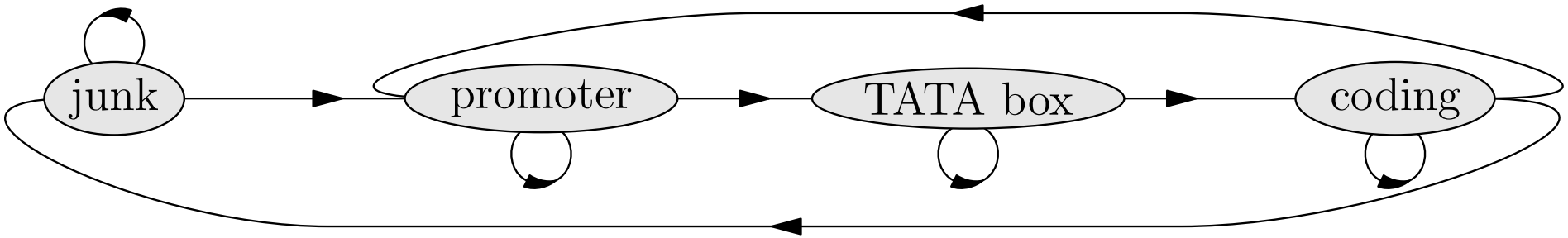

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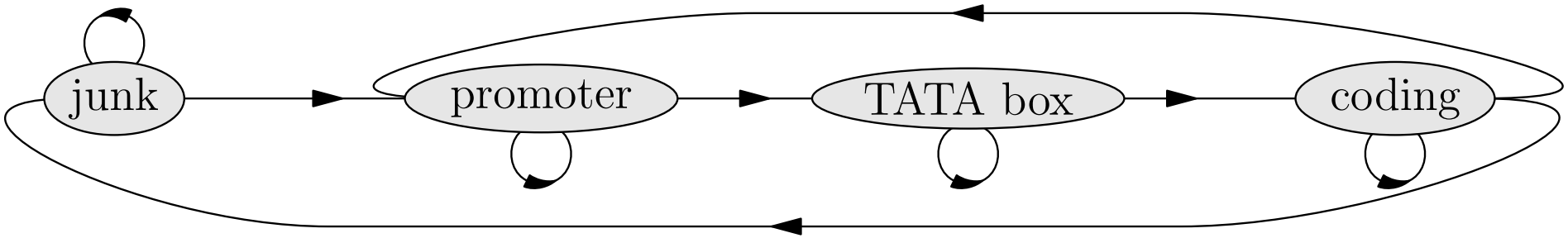

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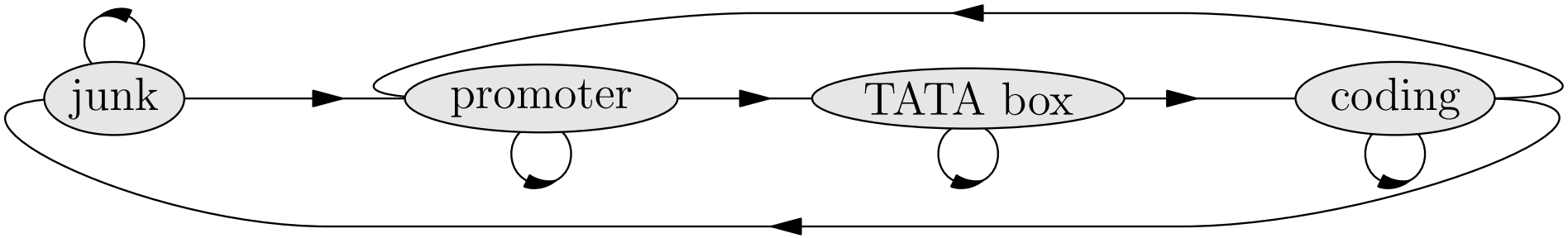

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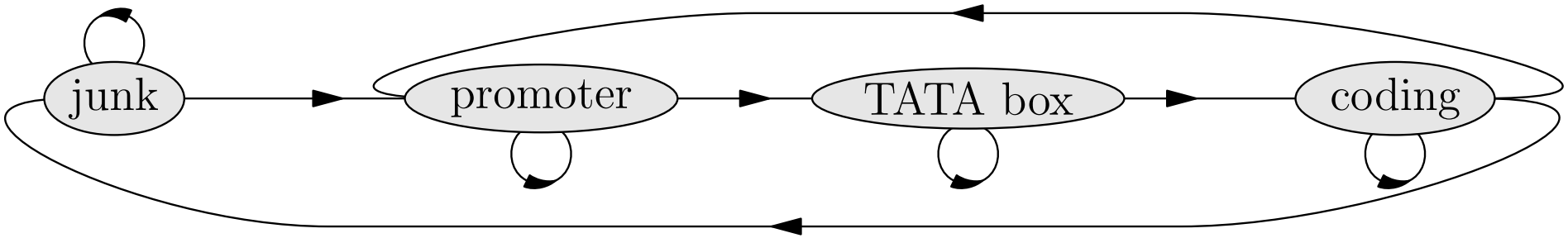

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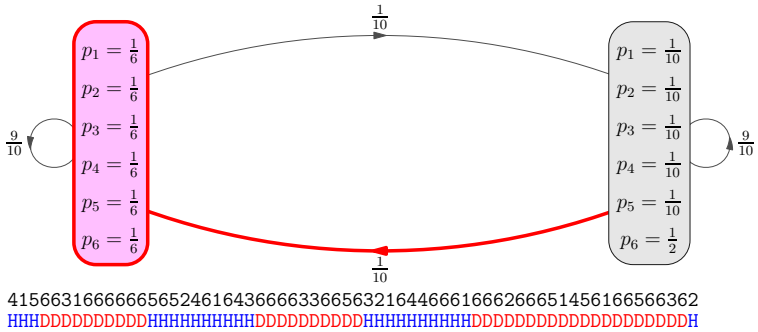

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- Although we have a generative model we want to use it as a descriminative model
- The problem is we usually don't know which region of DNA we are in (that is often what we are trying to find out)
- Thus we have a sequence, but we don't know which state we're in
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# Dishonest Casino

- Consider a dishonest casino where we have two dice
- An honest dice with $p_i = \frac{1}{6}$
- And a dishonest dice where $p_6 = \frac{1}{2}$ while $p_i = \frac{1}{10}$ for $i = \{1, 2, 3, 4, 5\}$
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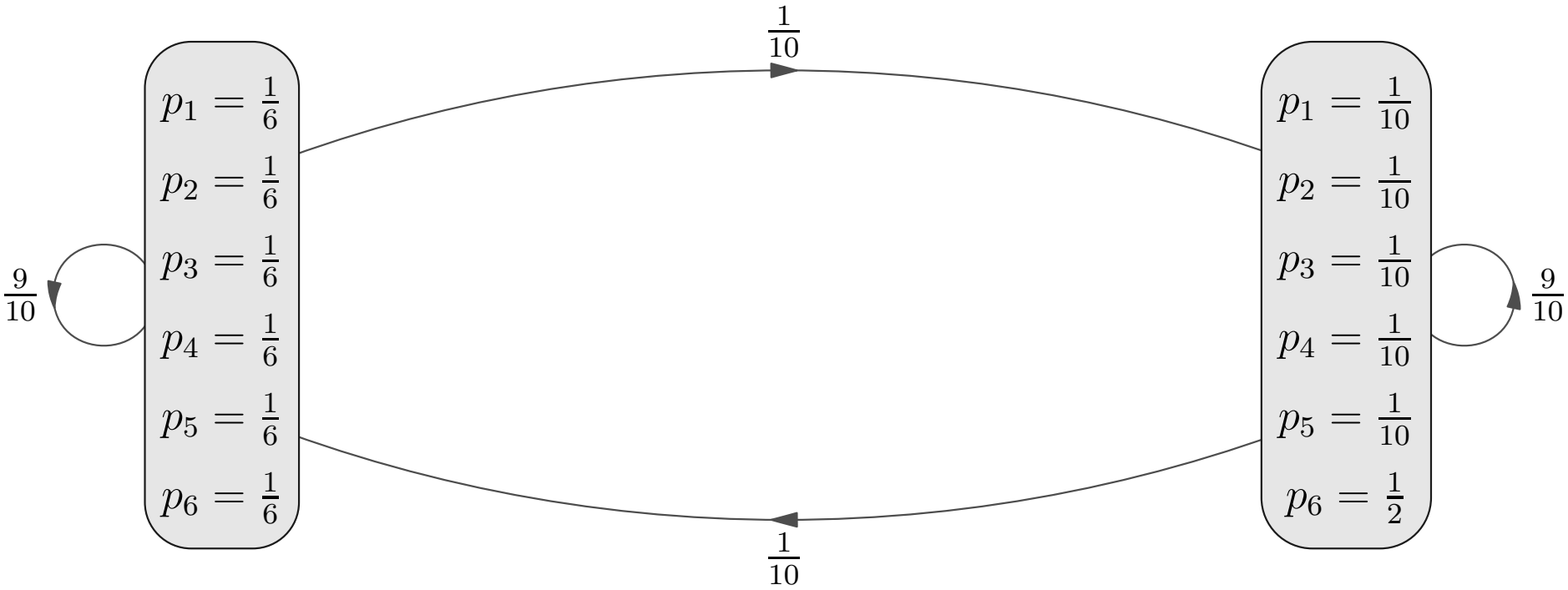

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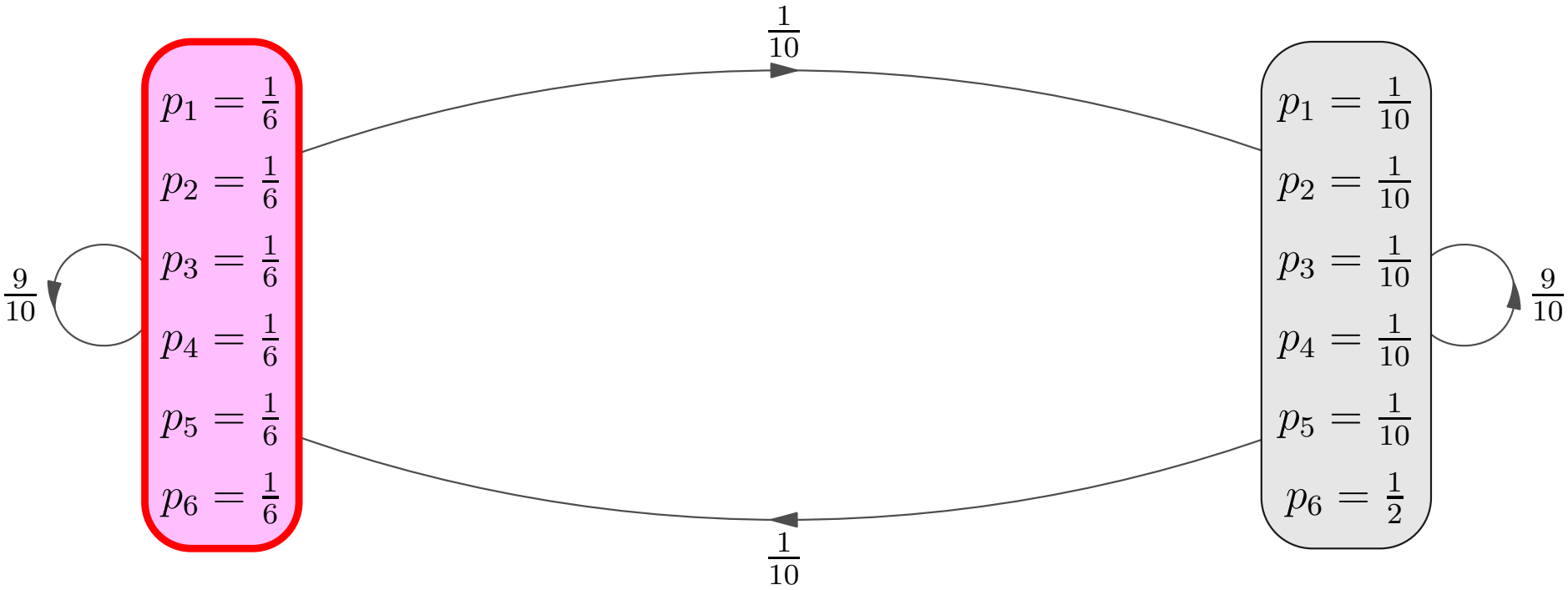


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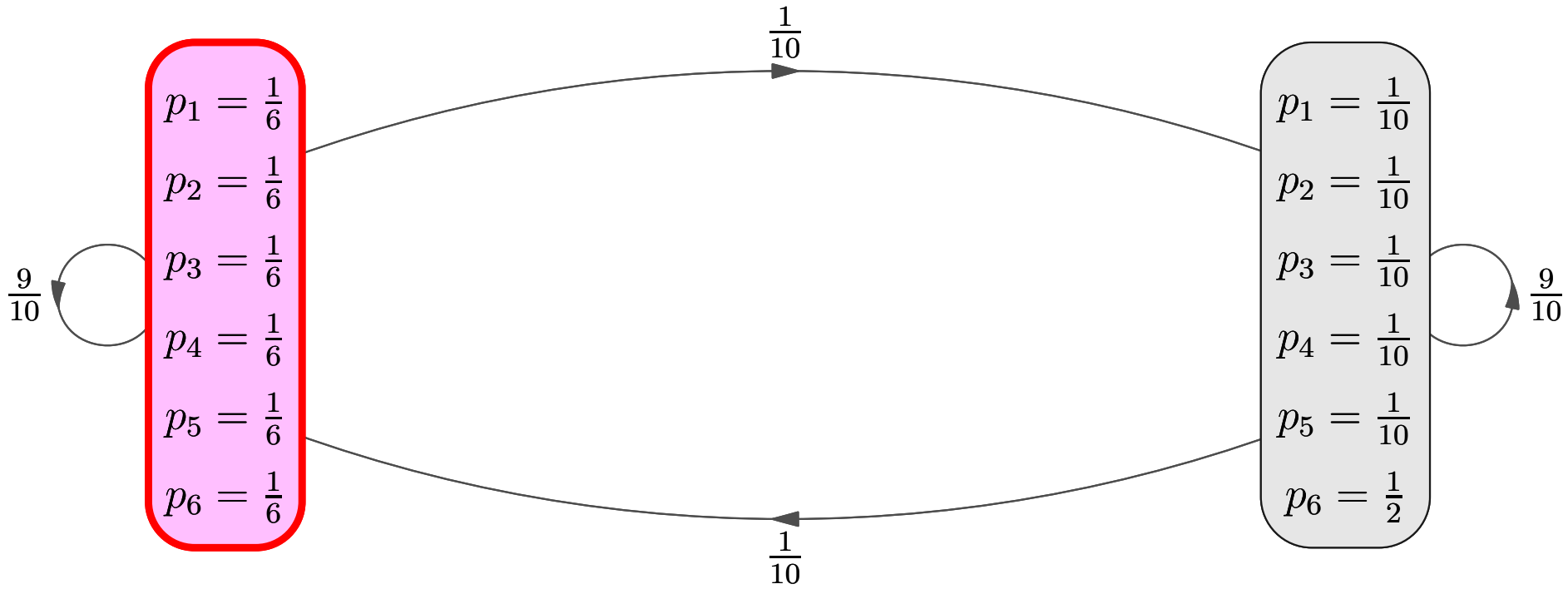


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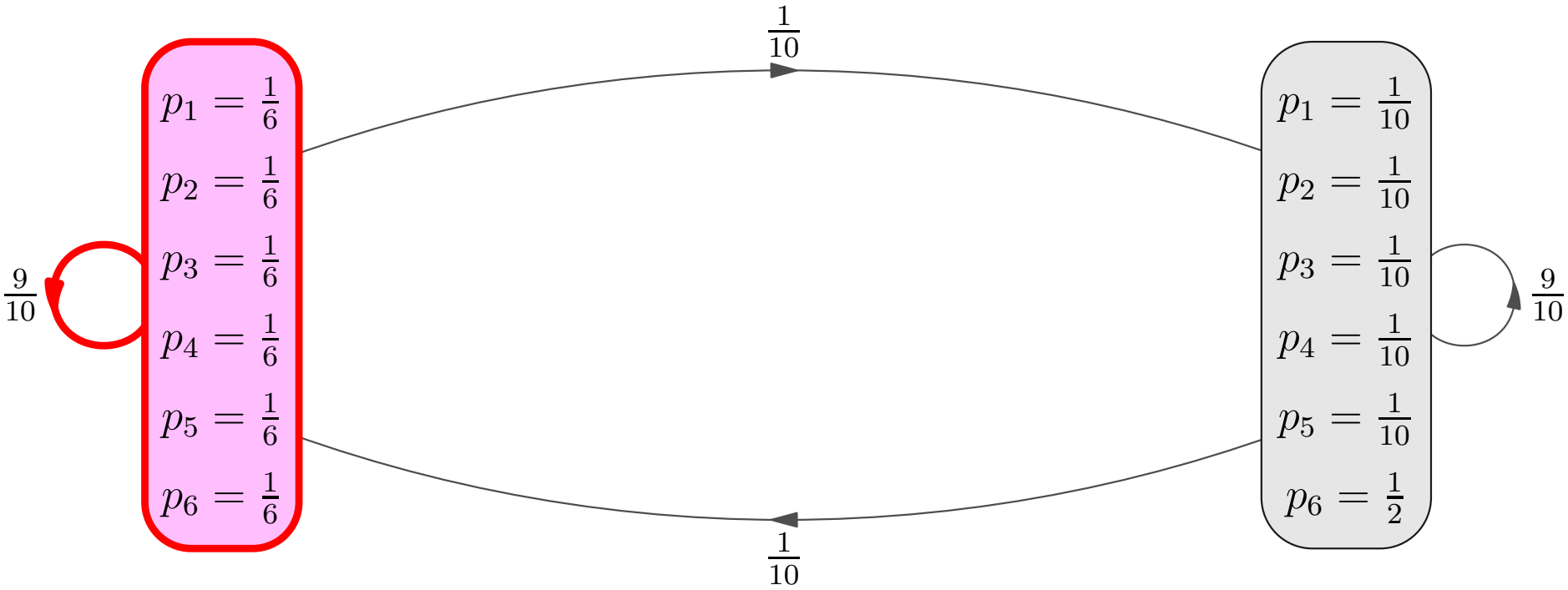


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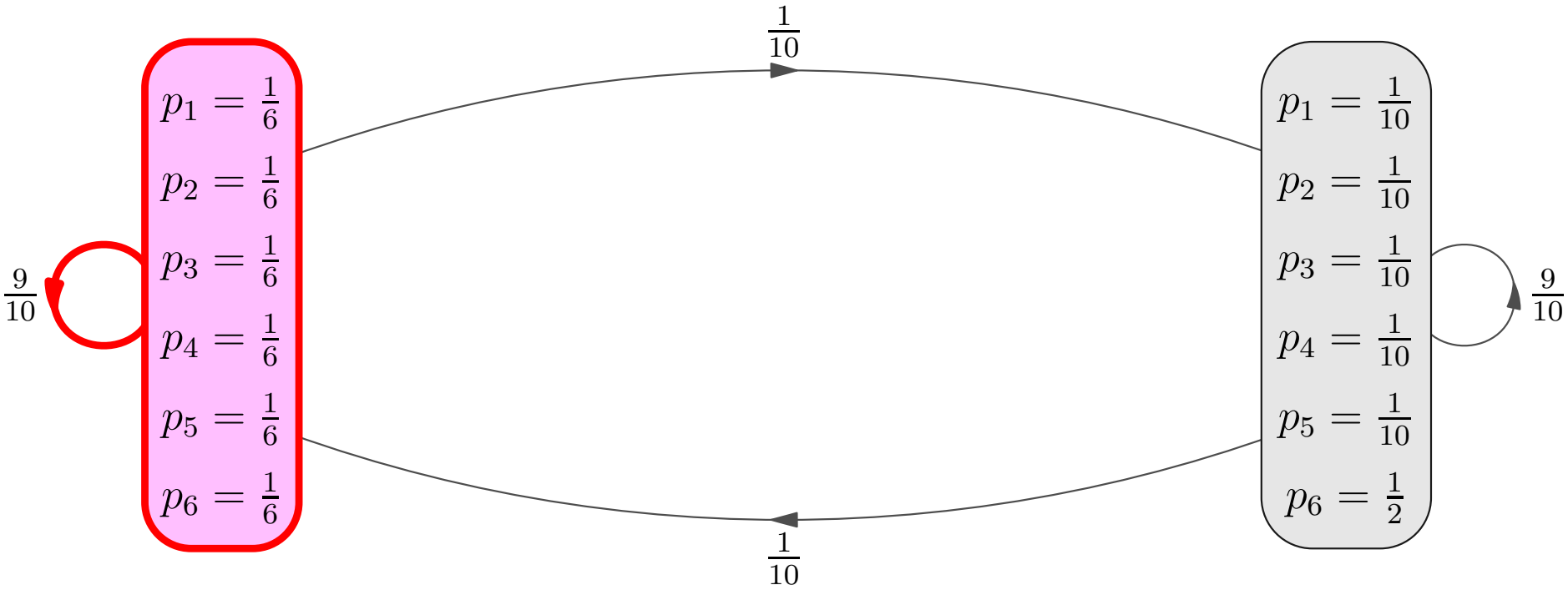


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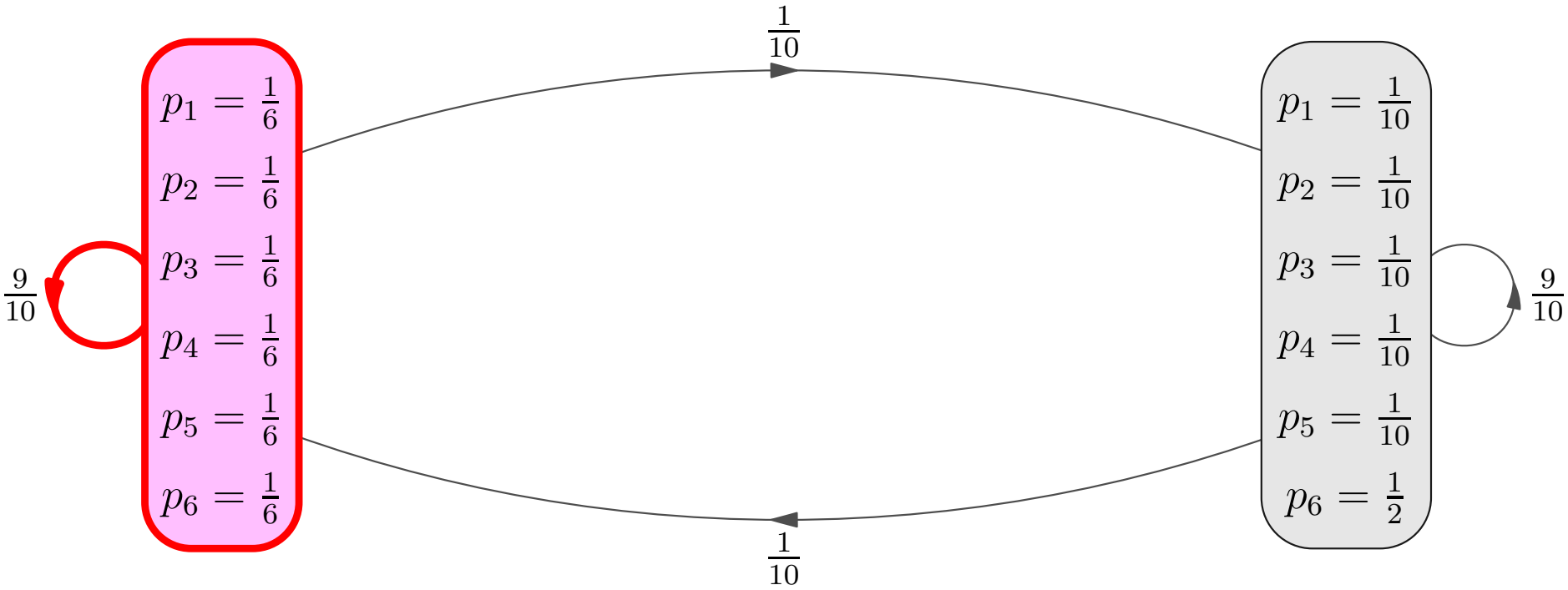


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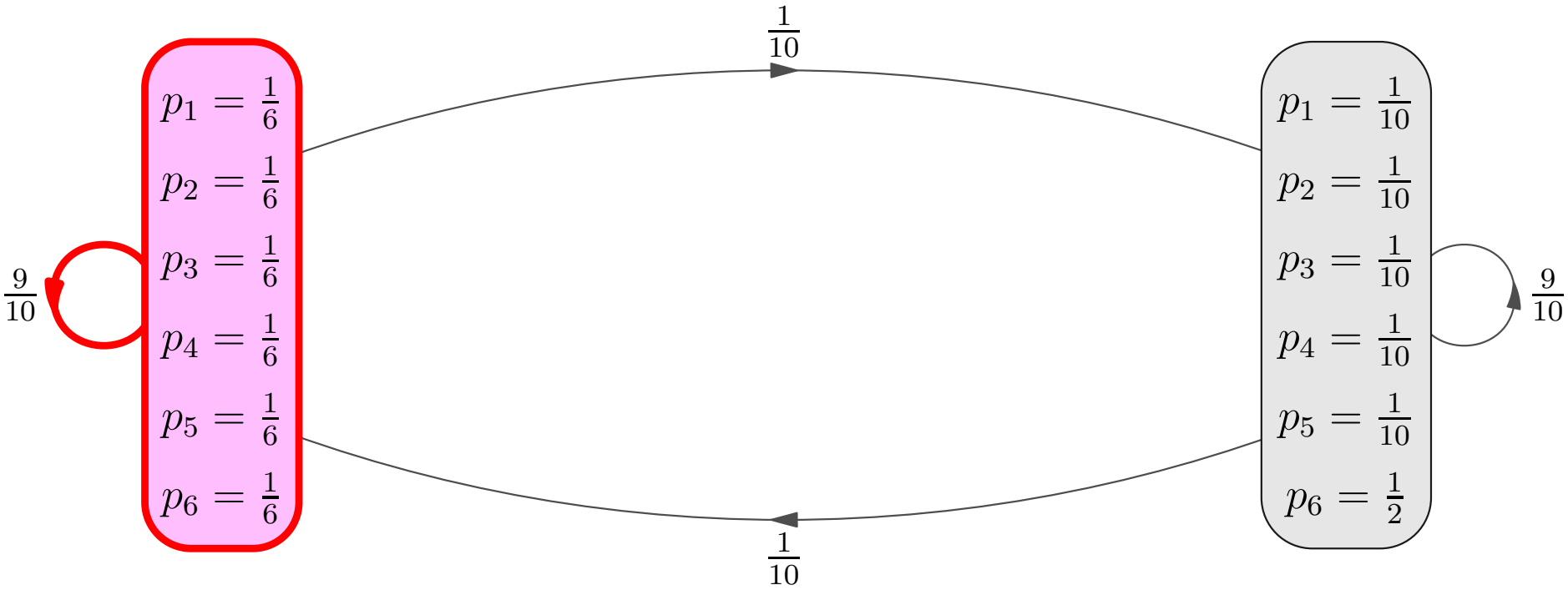


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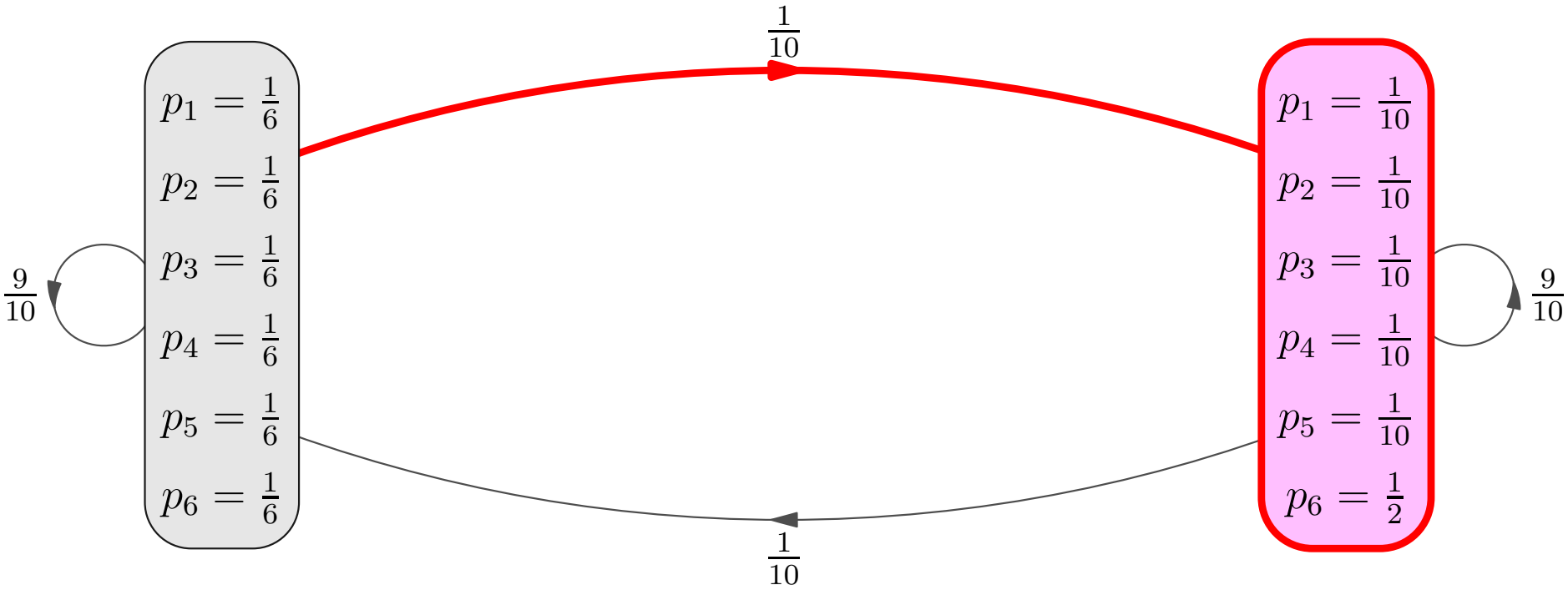


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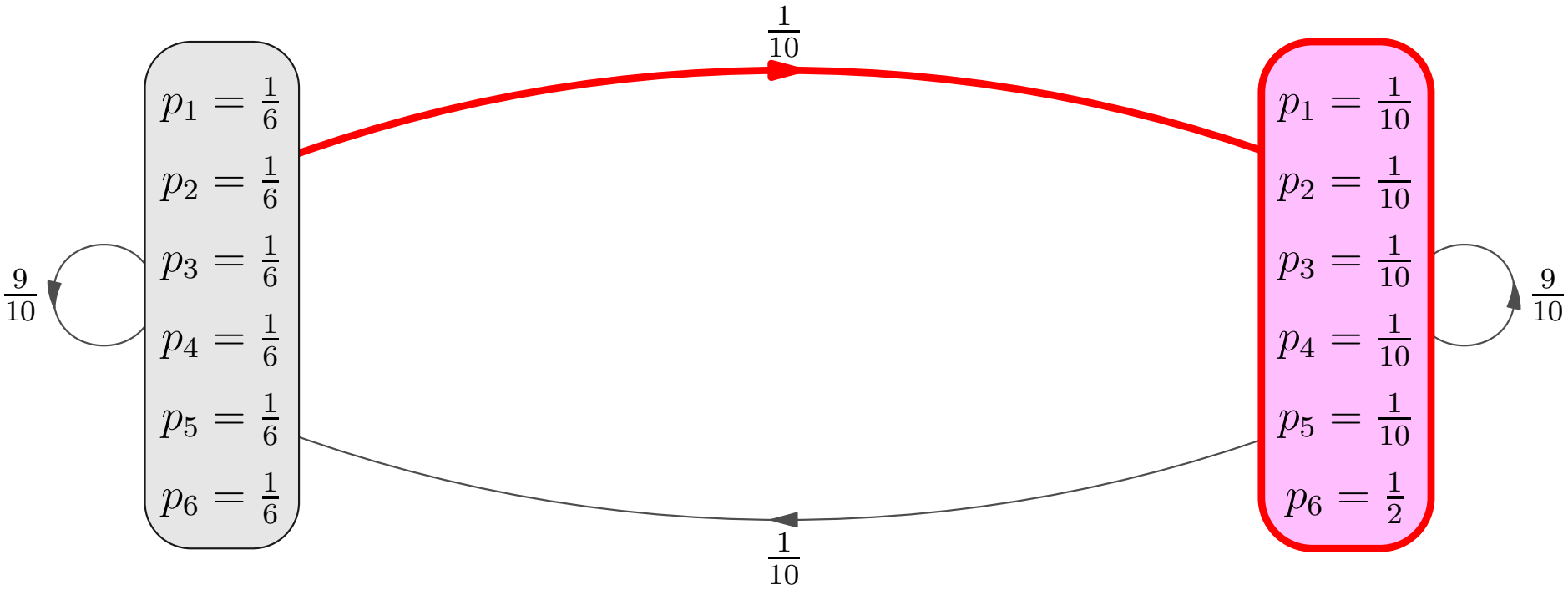


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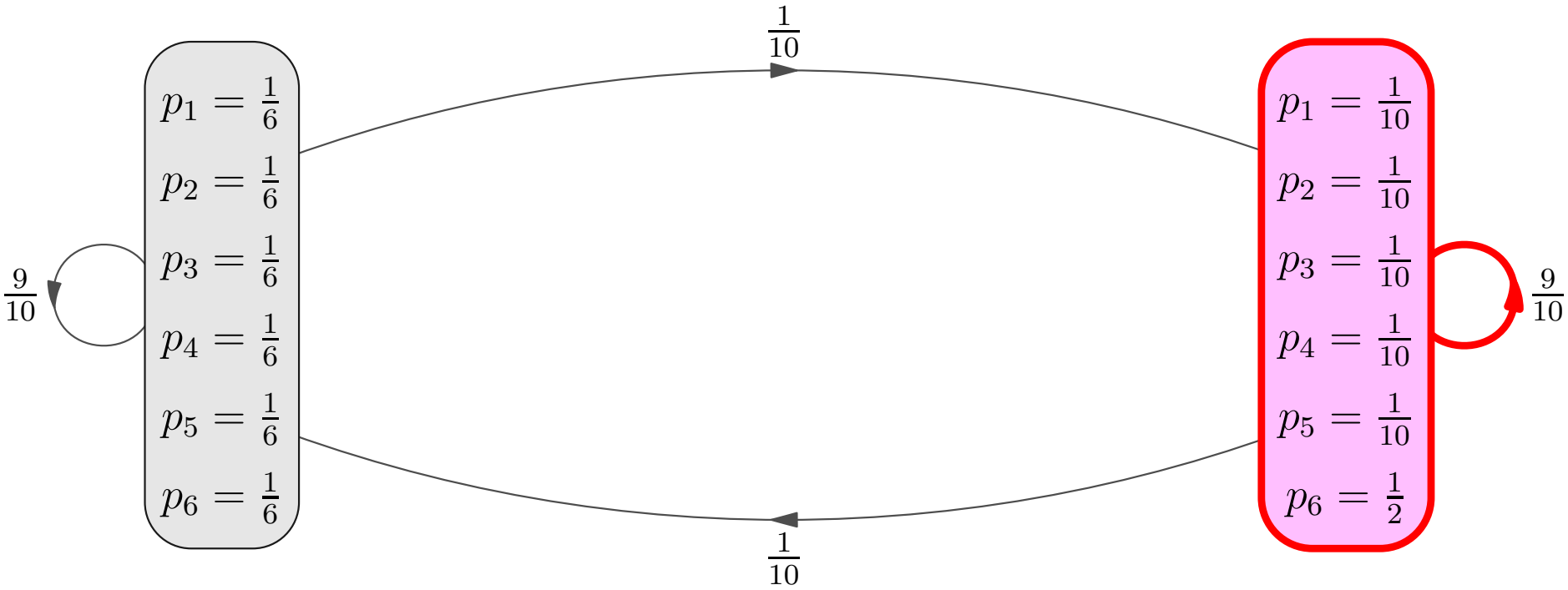


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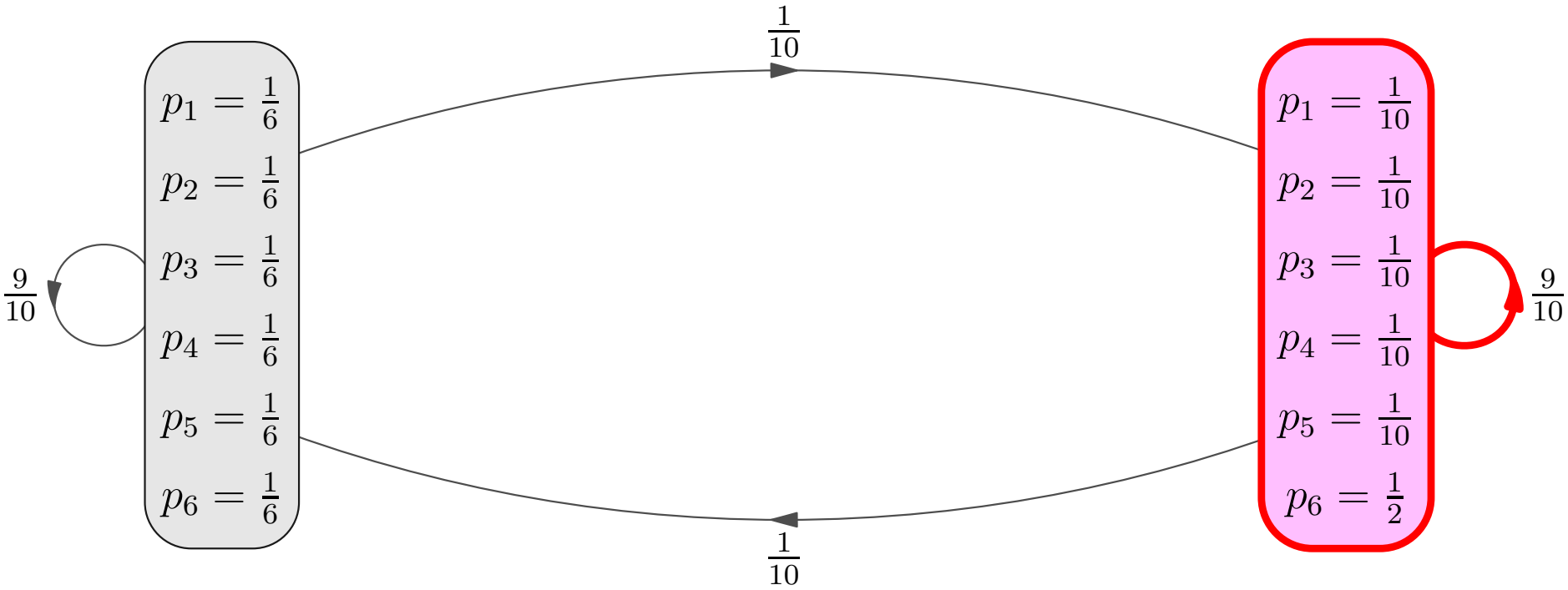


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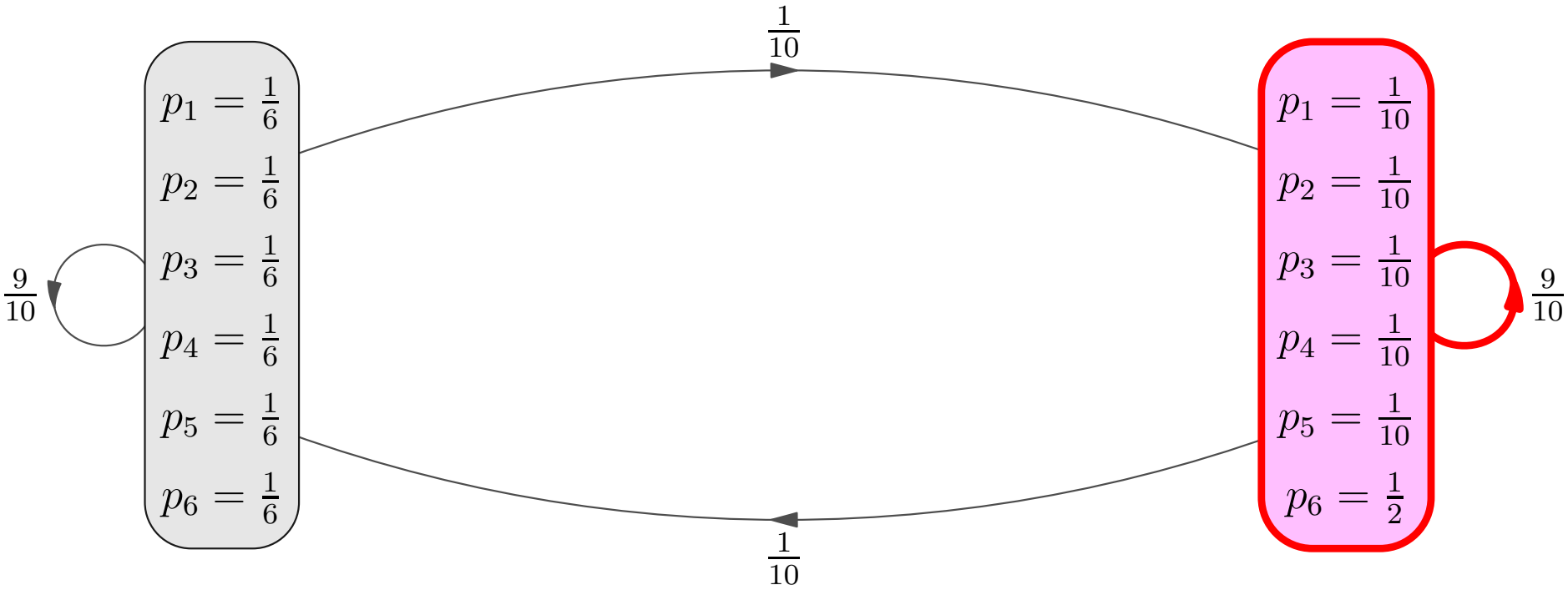


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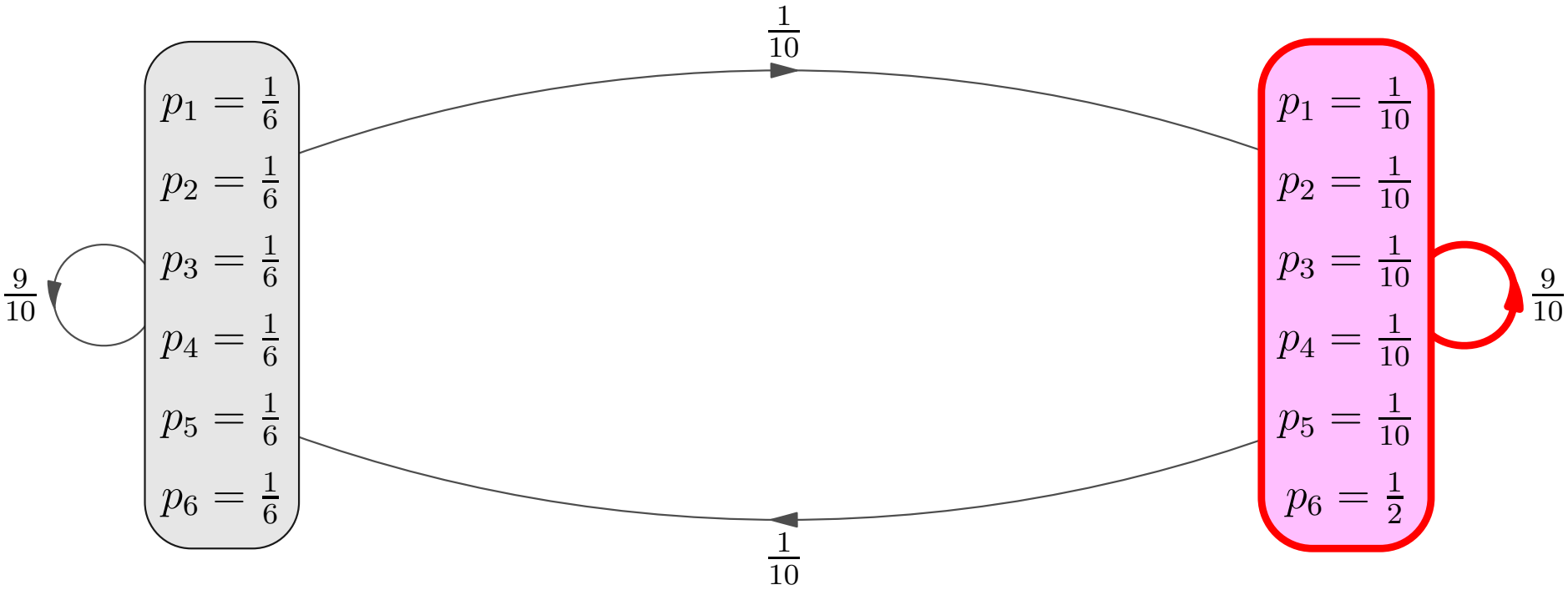


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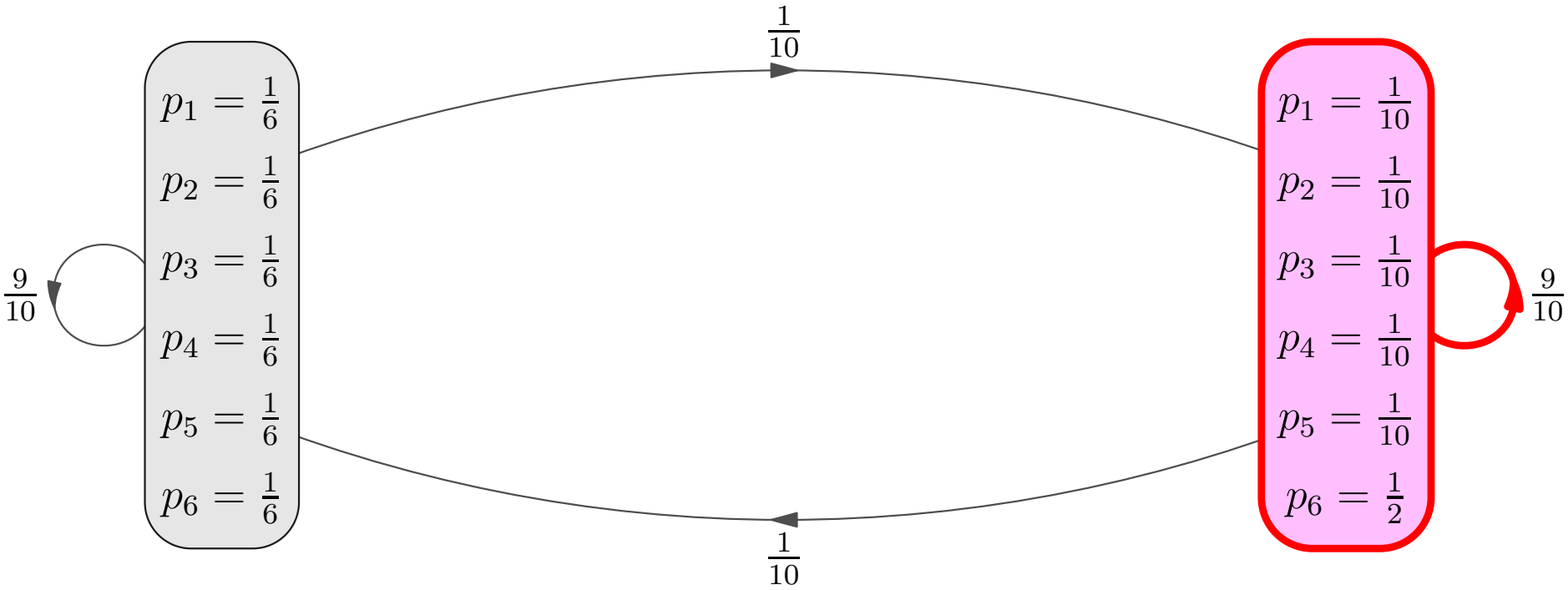


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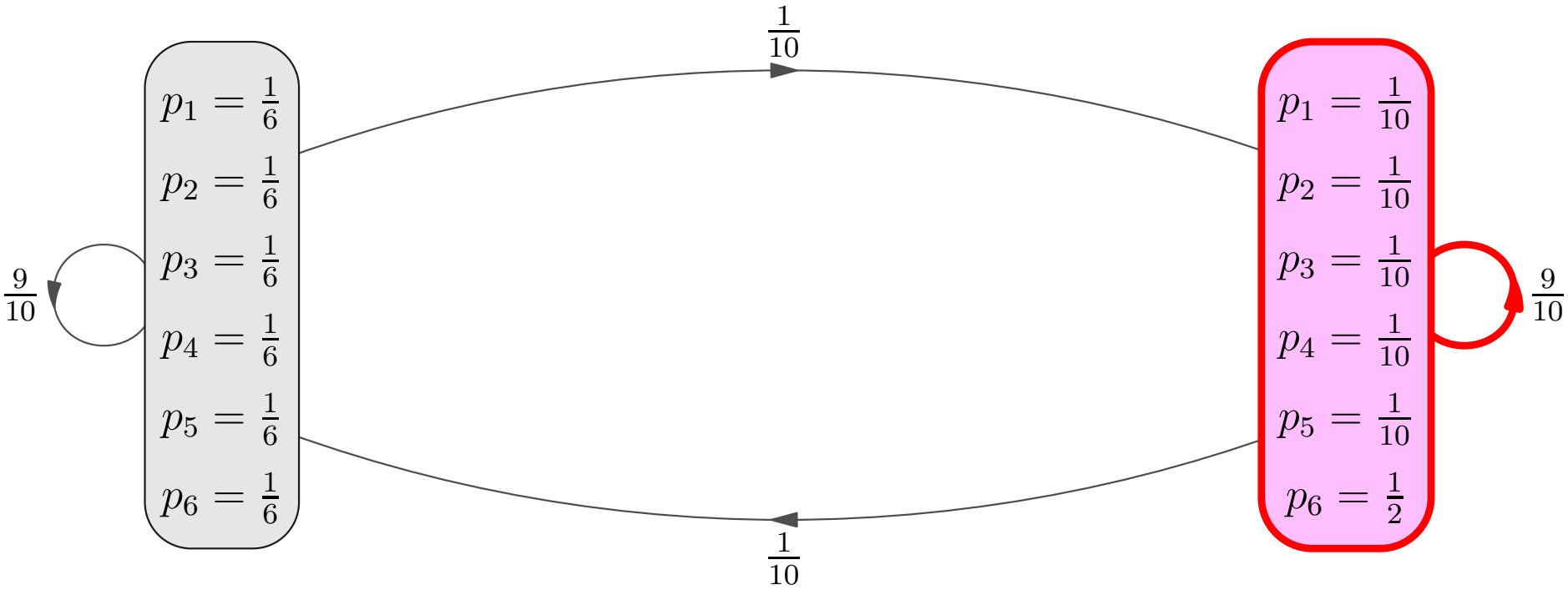


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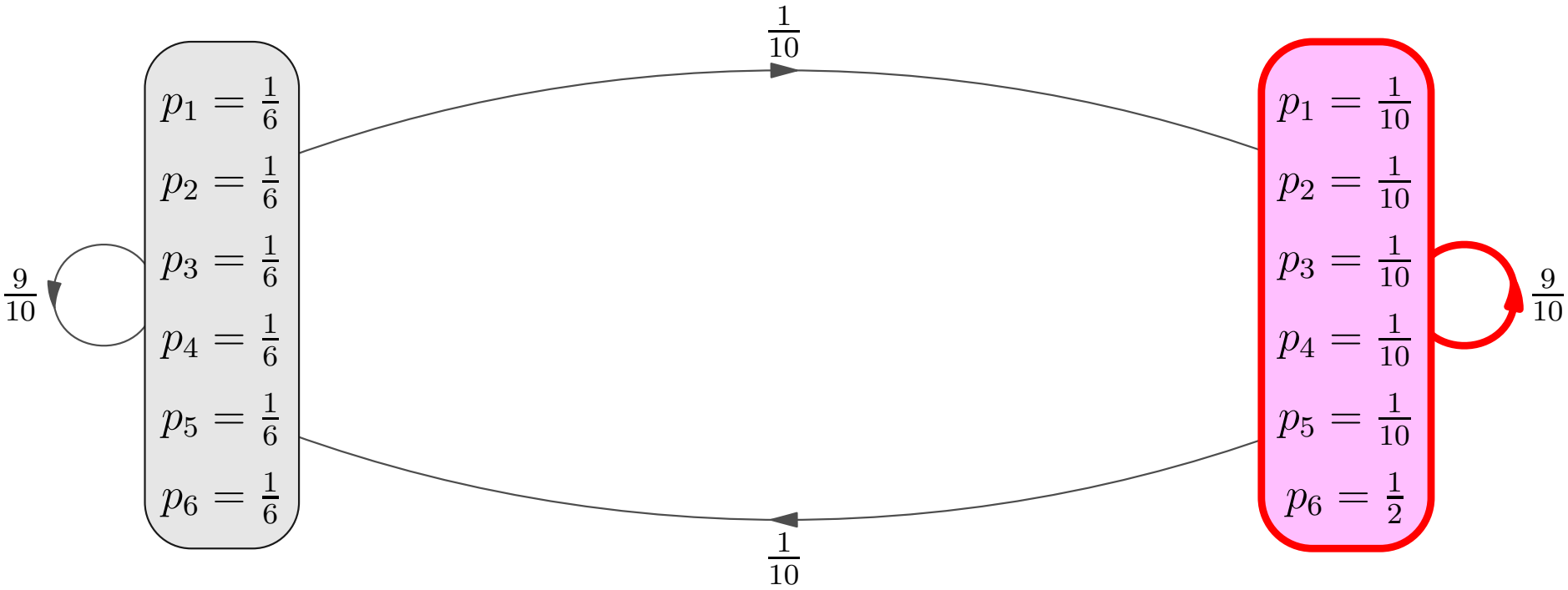


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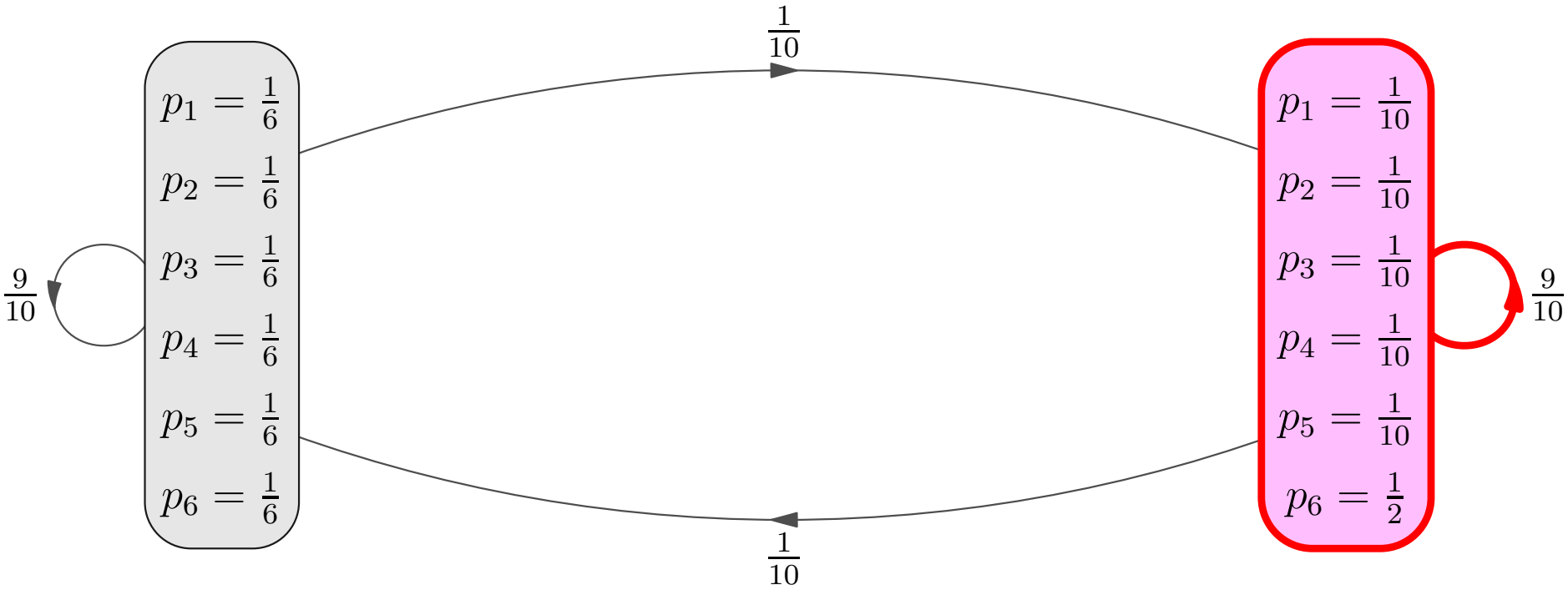


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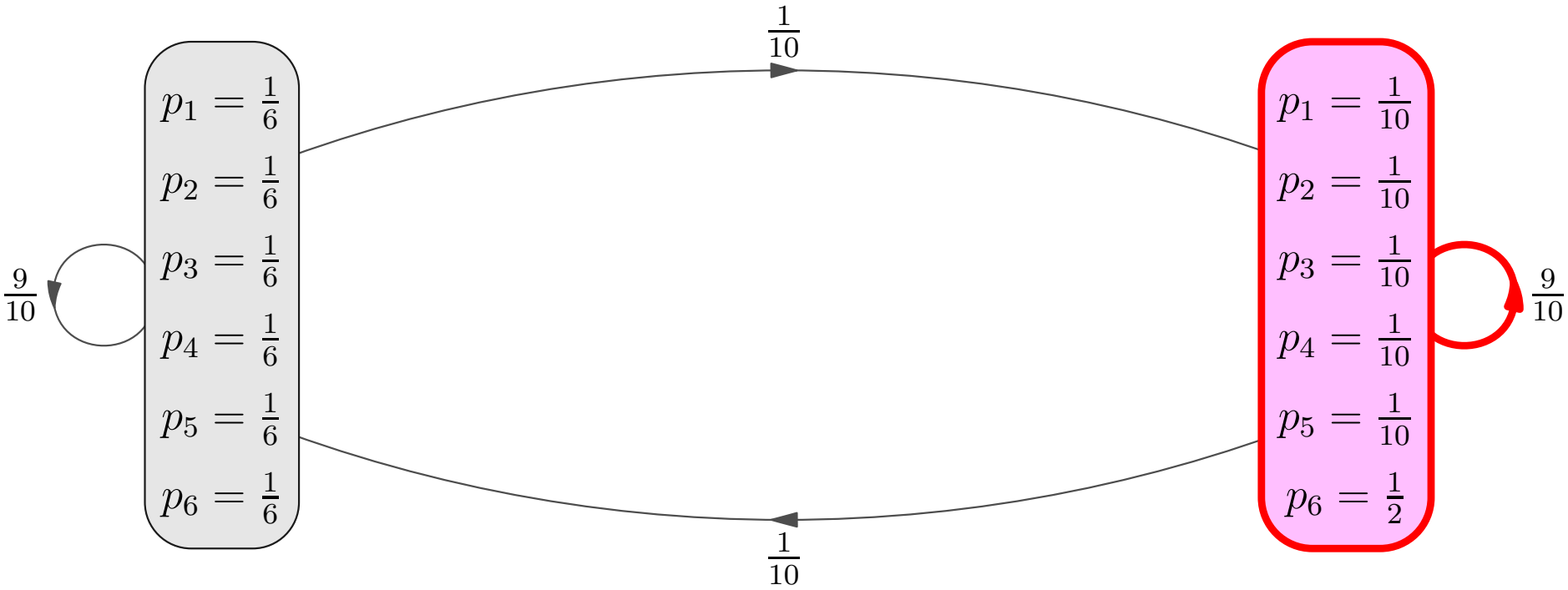


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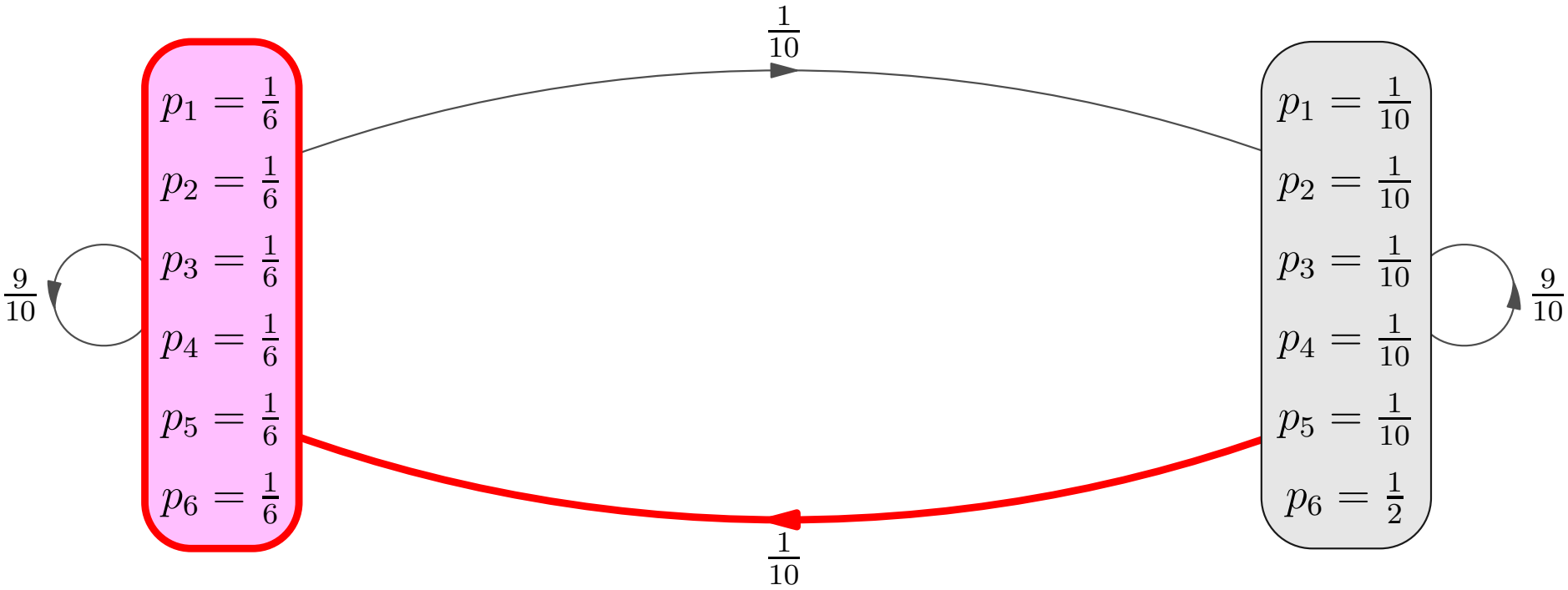


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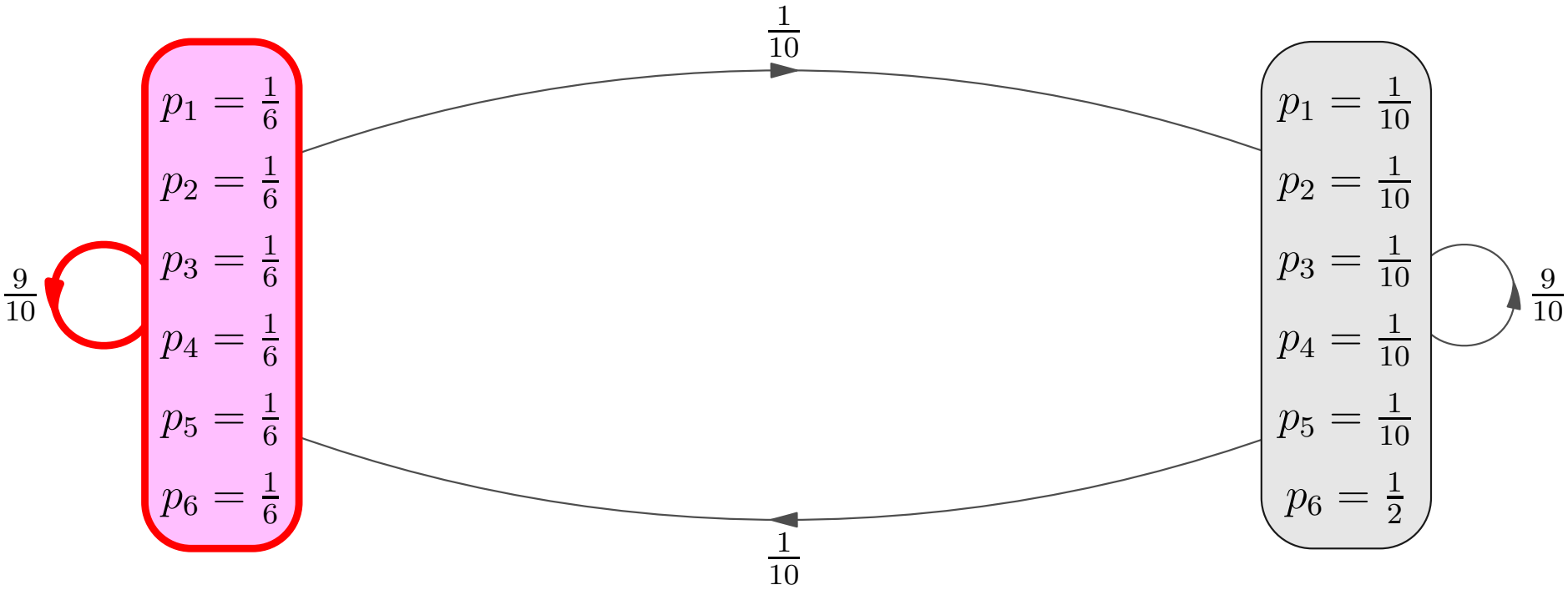


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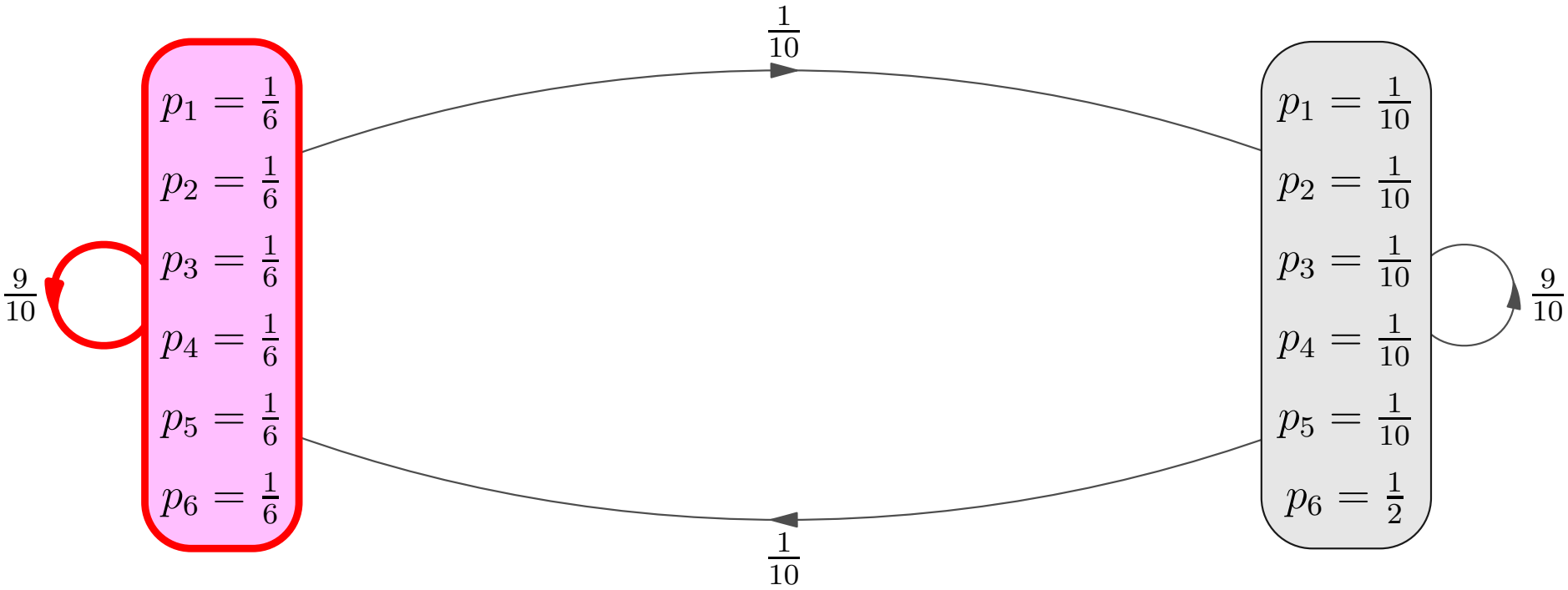


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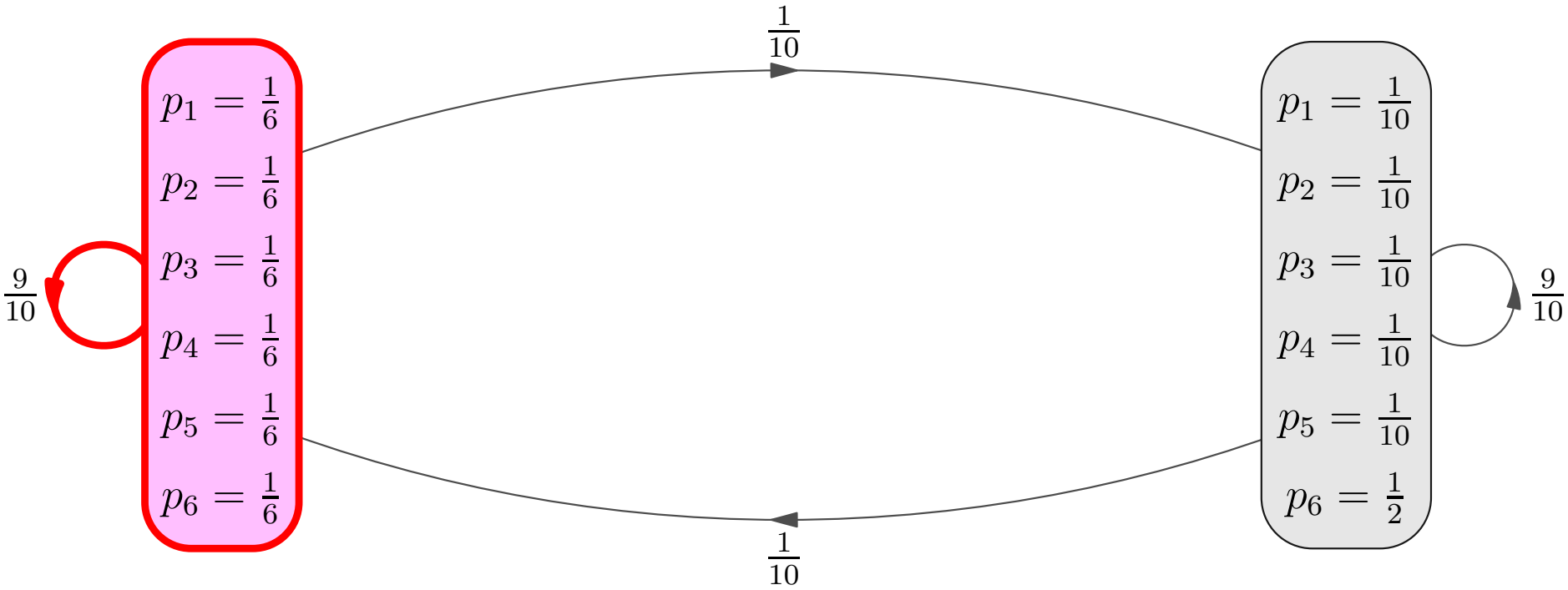


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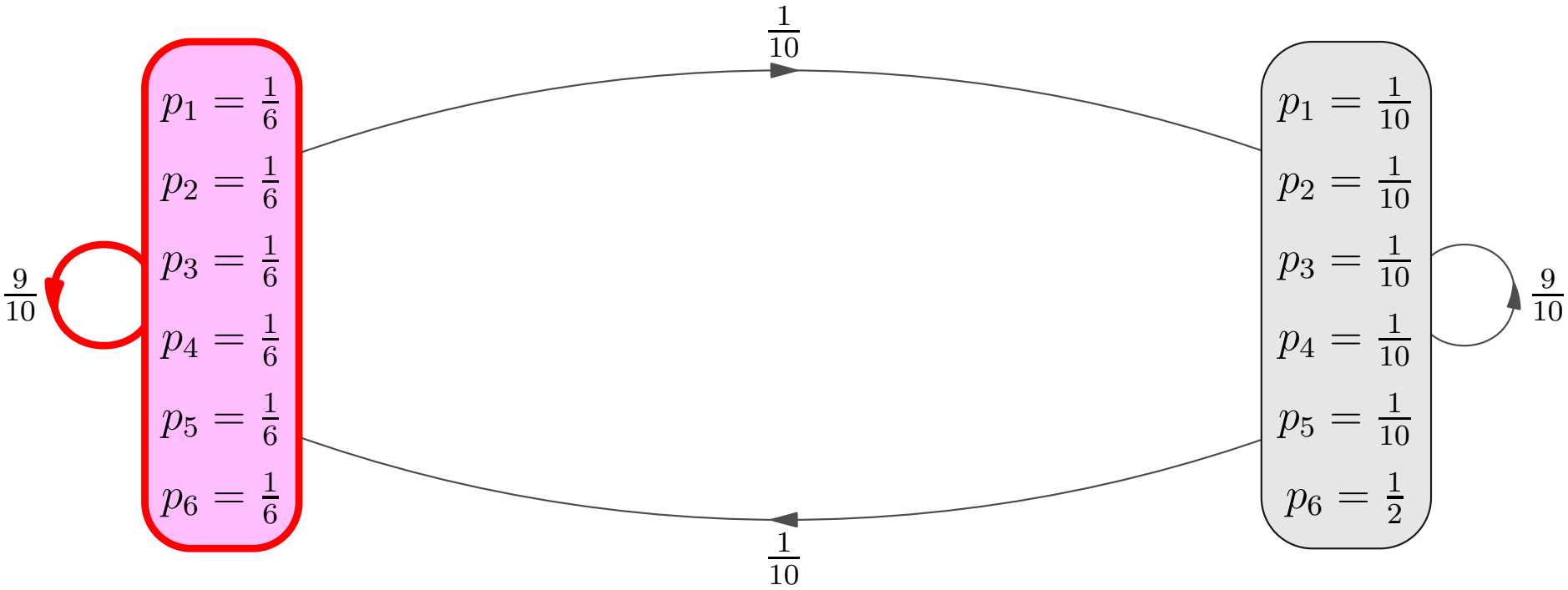


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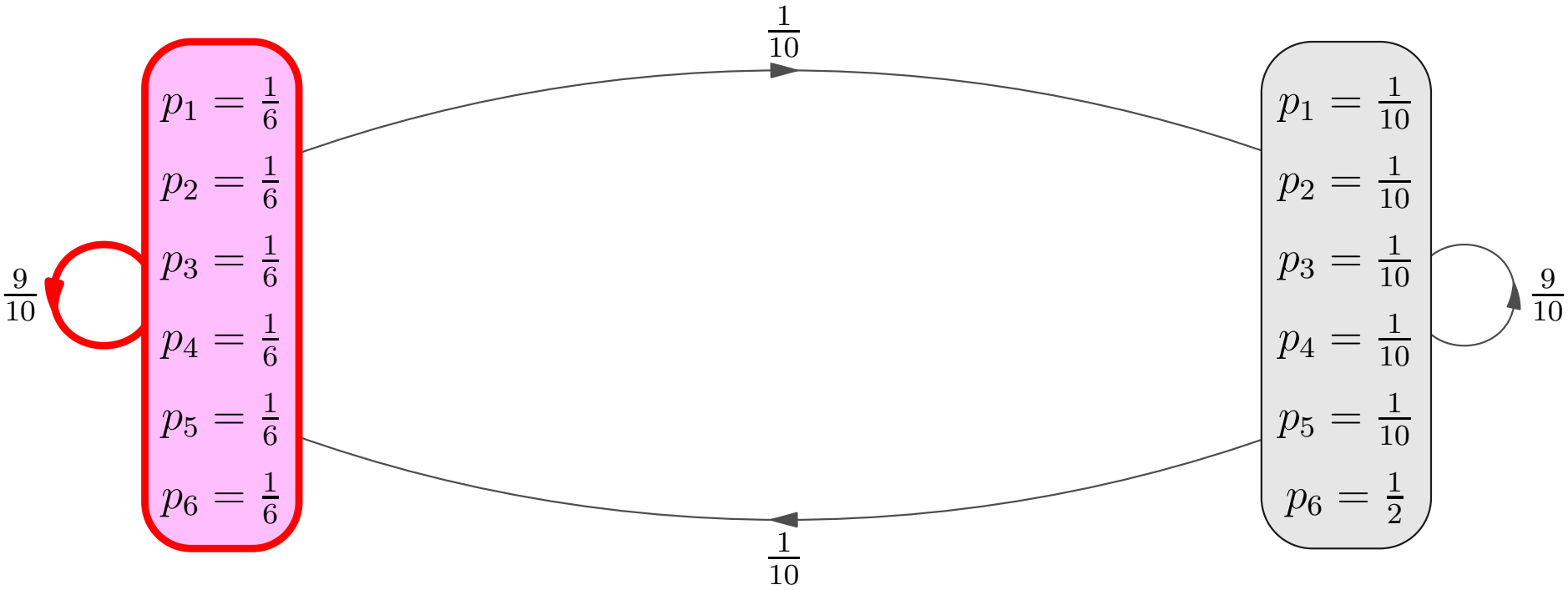


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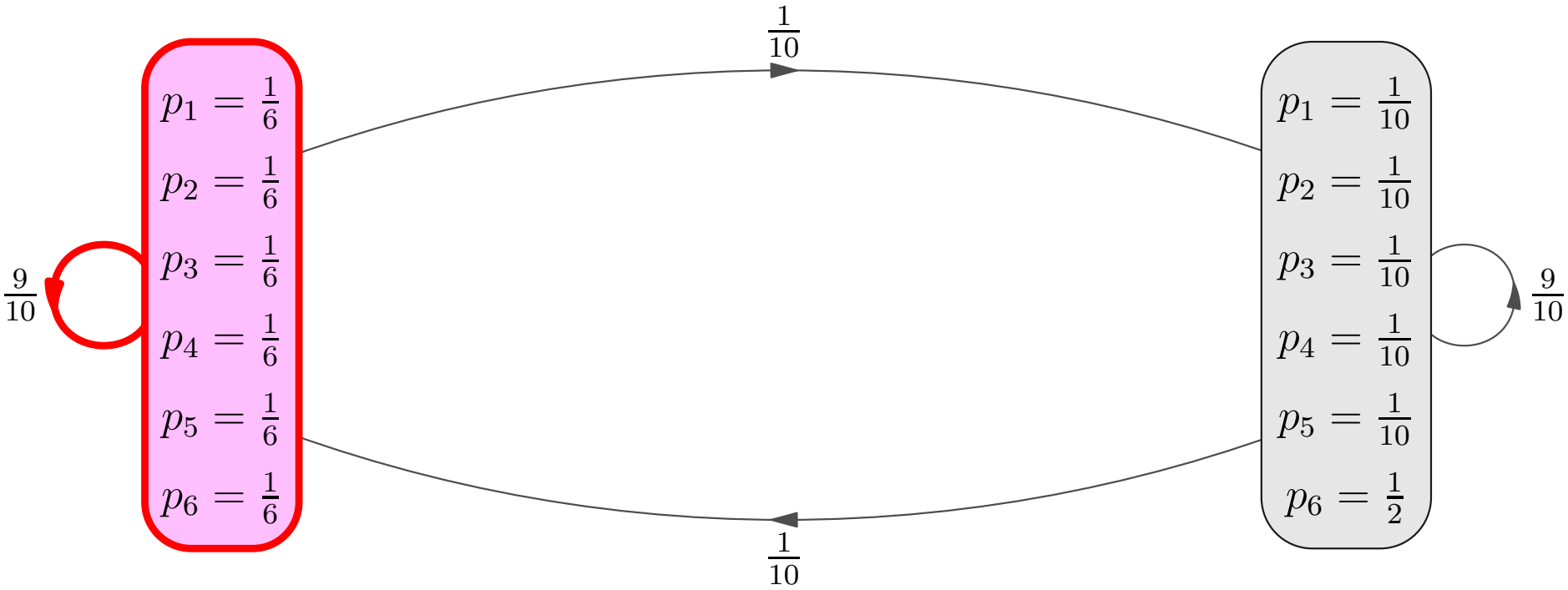


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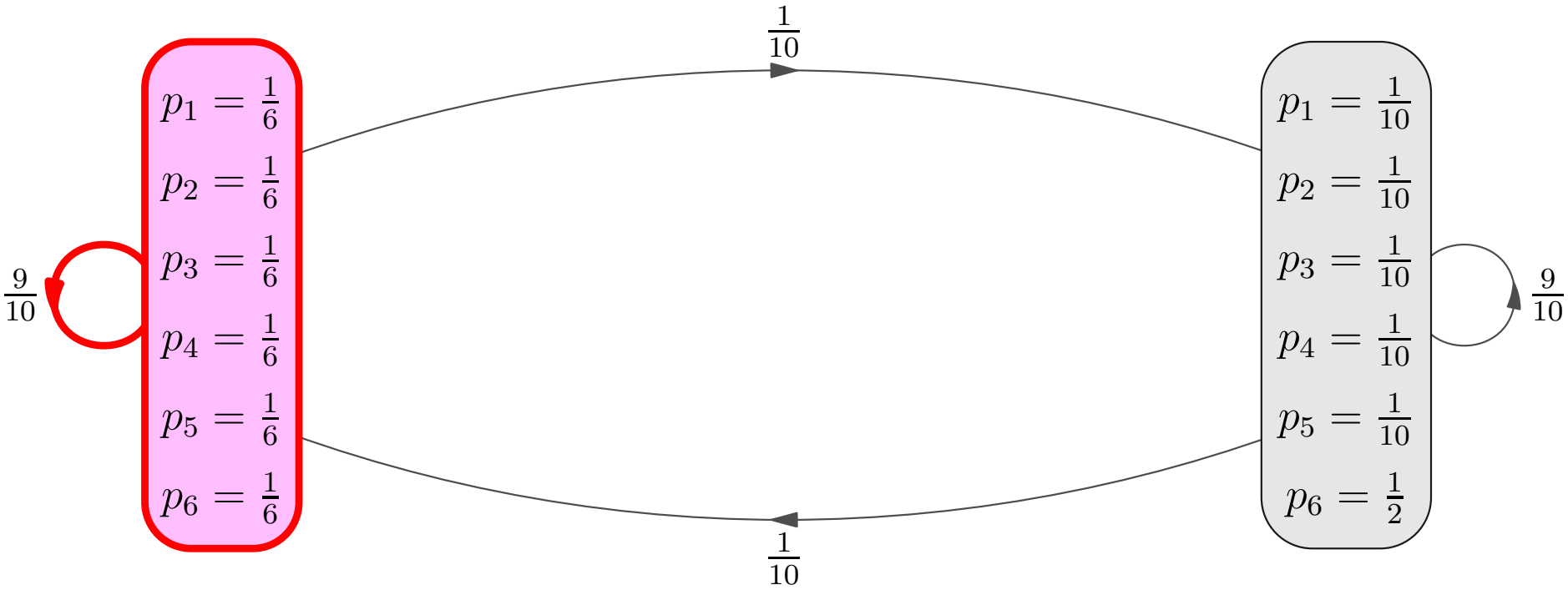


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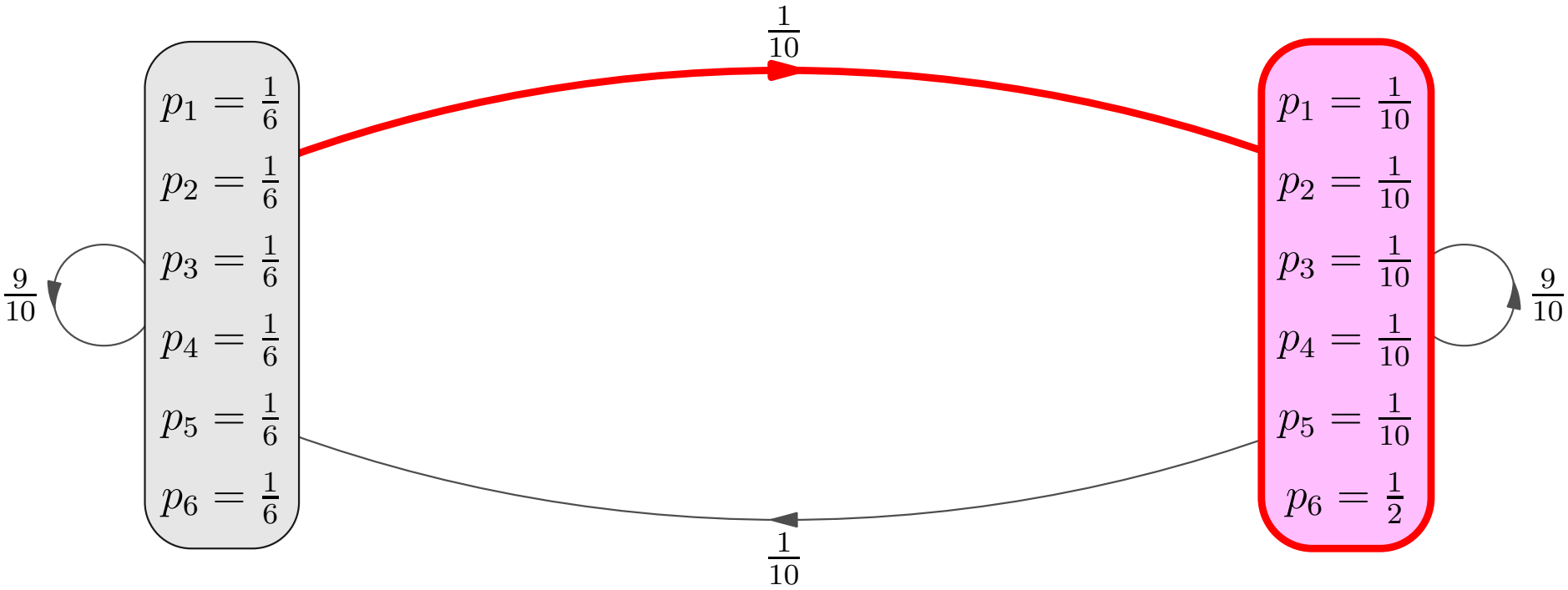


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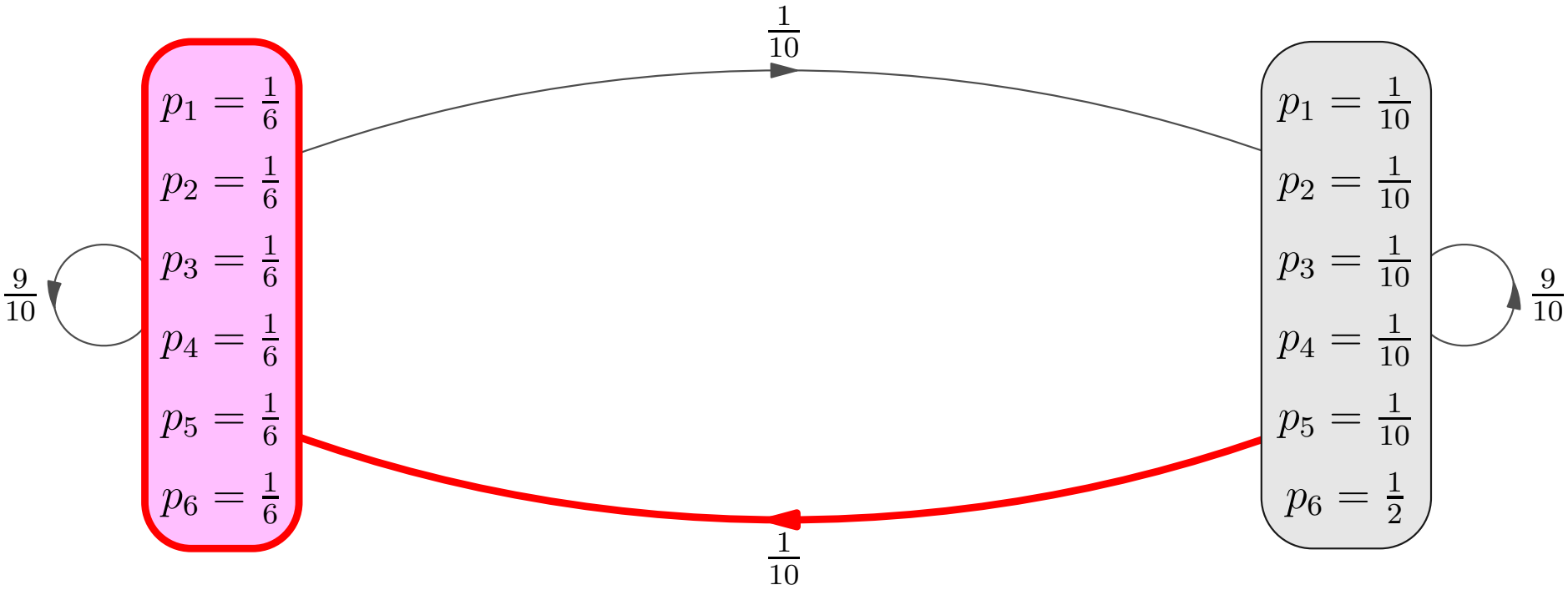


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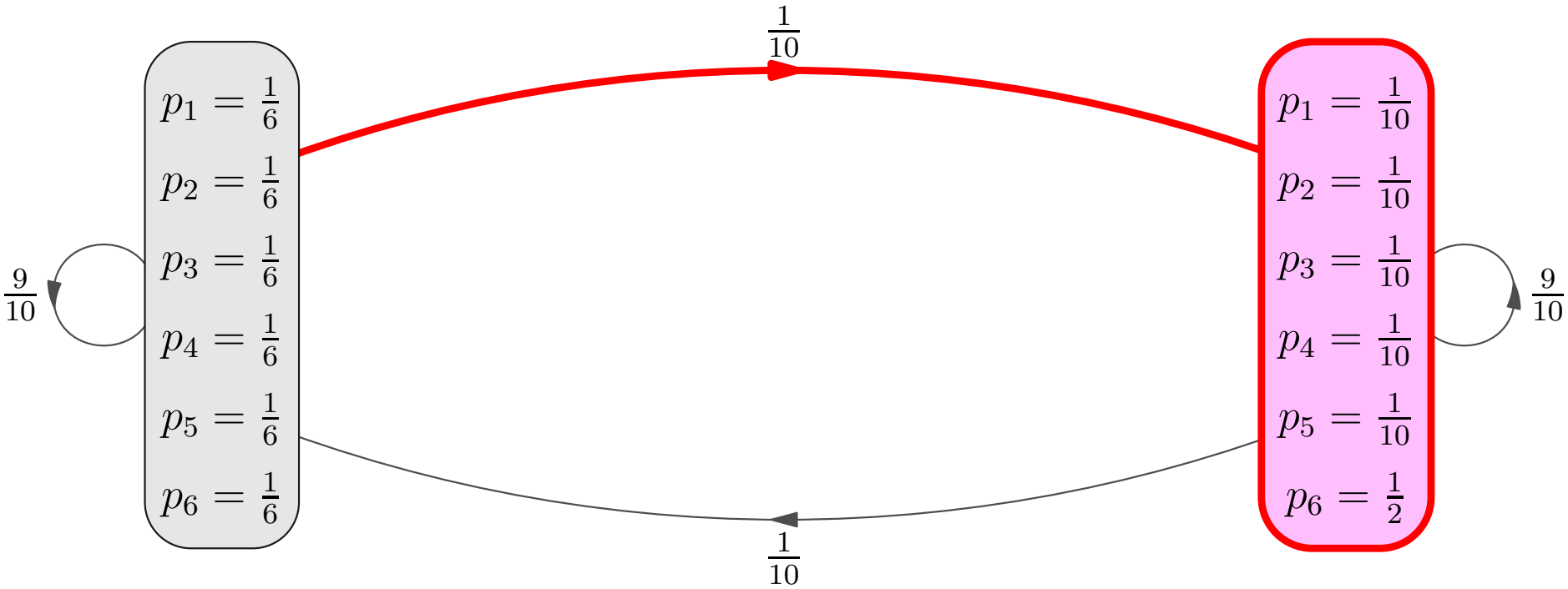


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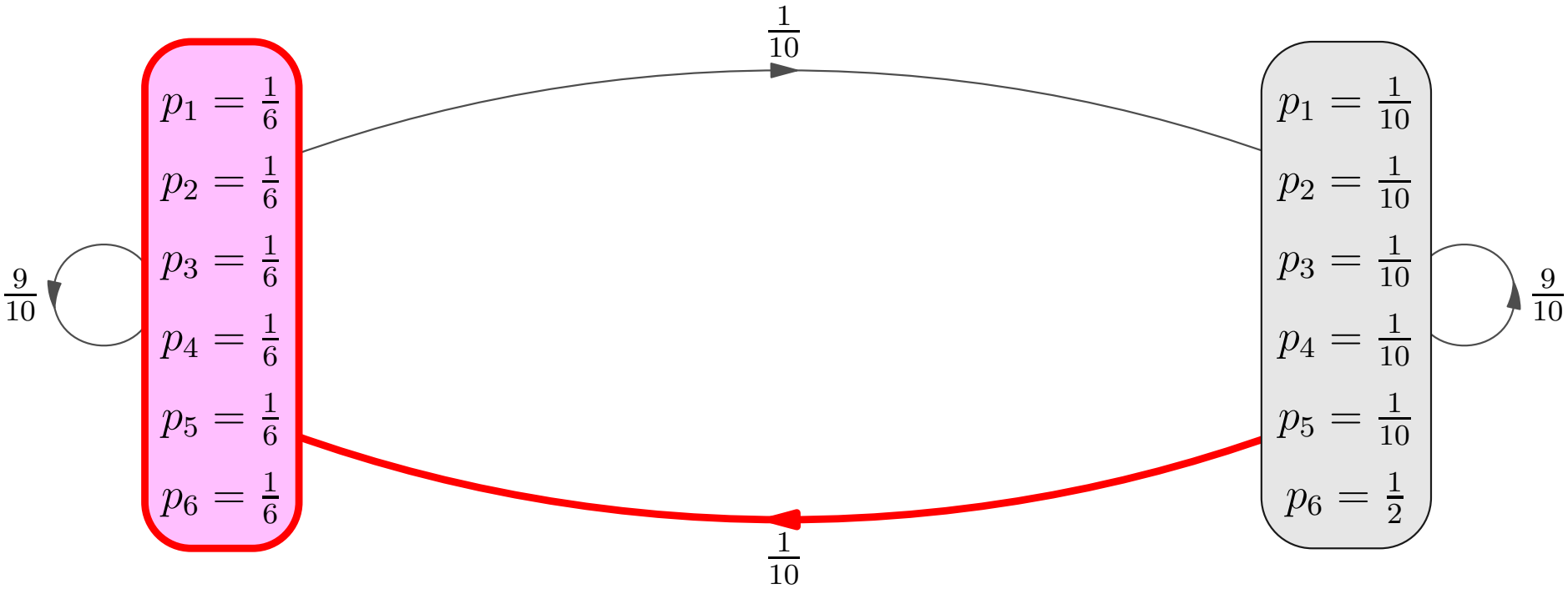


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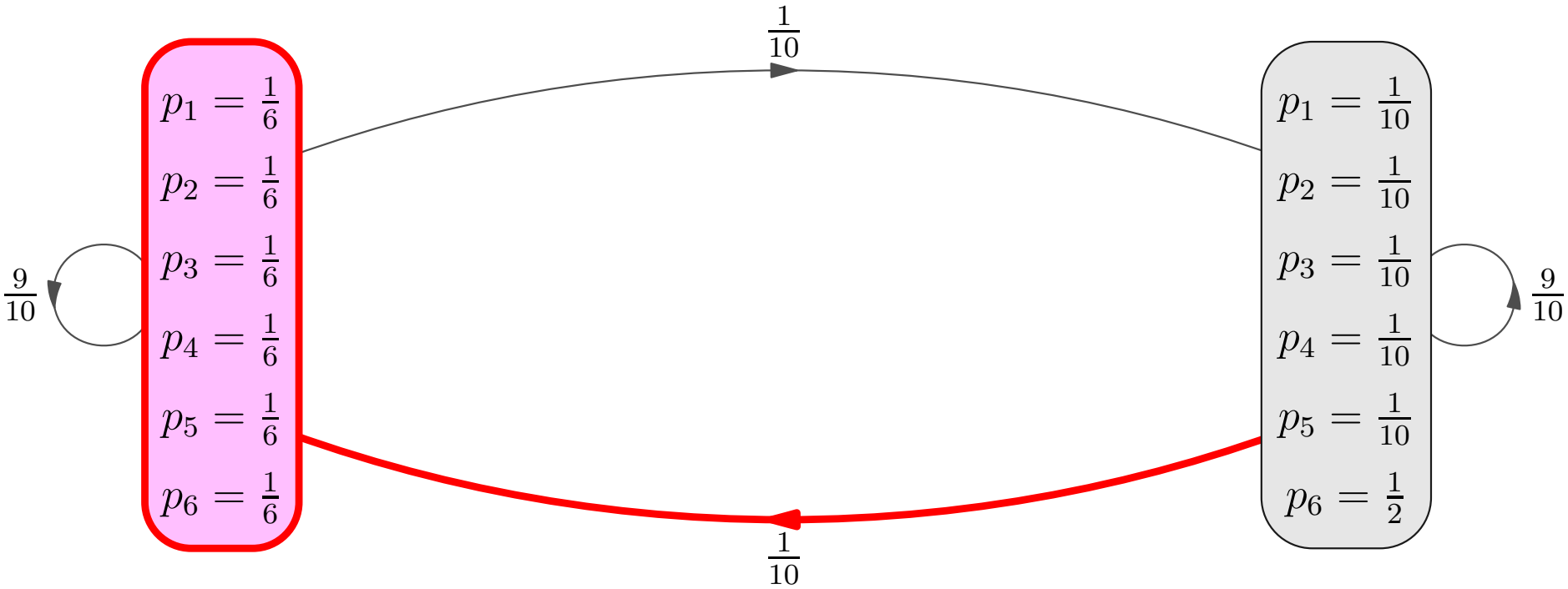


415663166666656524616436666336656321644666166626665145616656636 2
HHHDDDDDDDDDDHHHHHHHHHHHDDDDDDDDDDHHHHHHHHHHHDDDDDDDDDDDDDDDDDDDDDH

# HMM Terminology

- We consider a sequence consisting of letters $\zeta$ from some alphabet $\mathcal{A}$
- We denote our sequence by $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_T)$
- We let the set of states be denote by $\mathcal{Q}$
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- In HMMs we often have an initial state which we transition out of before doing anything else

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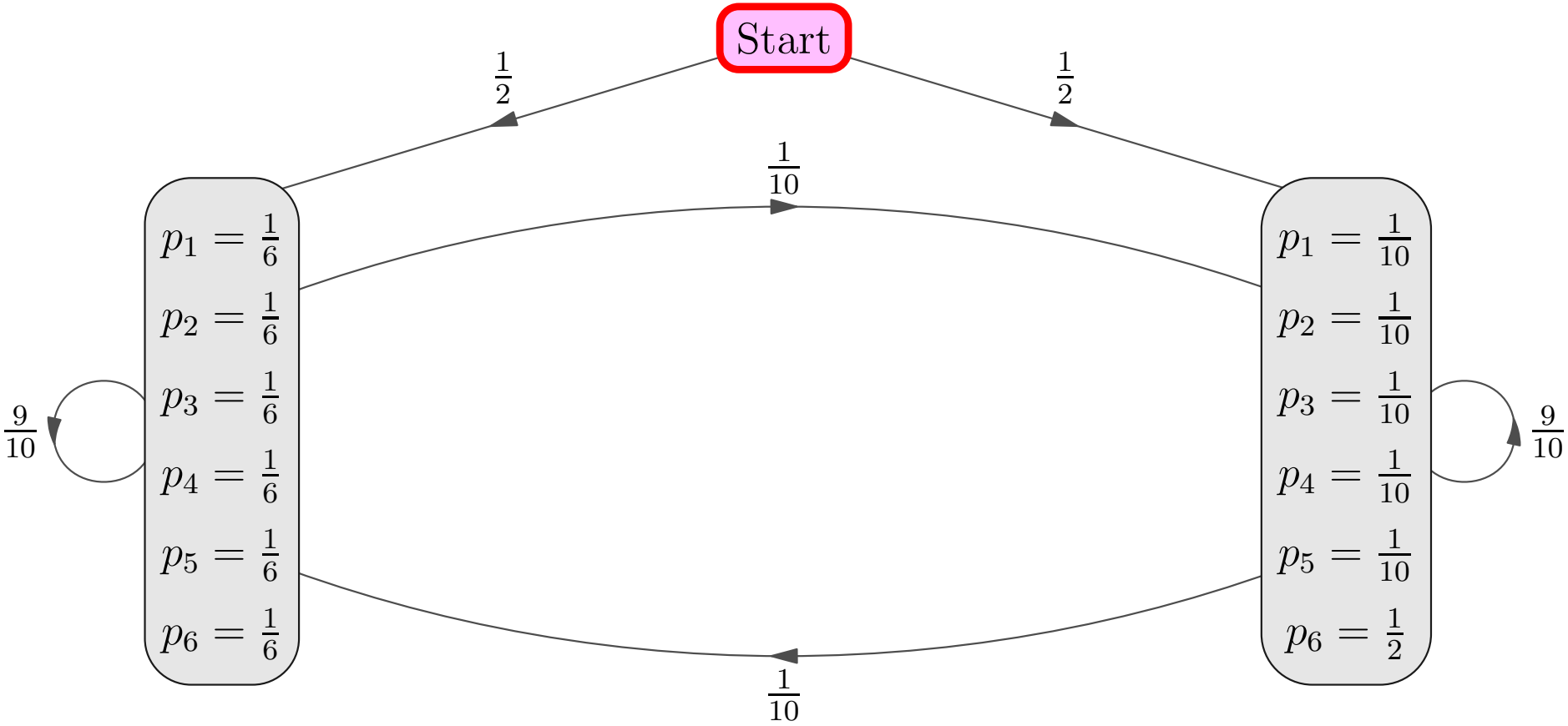
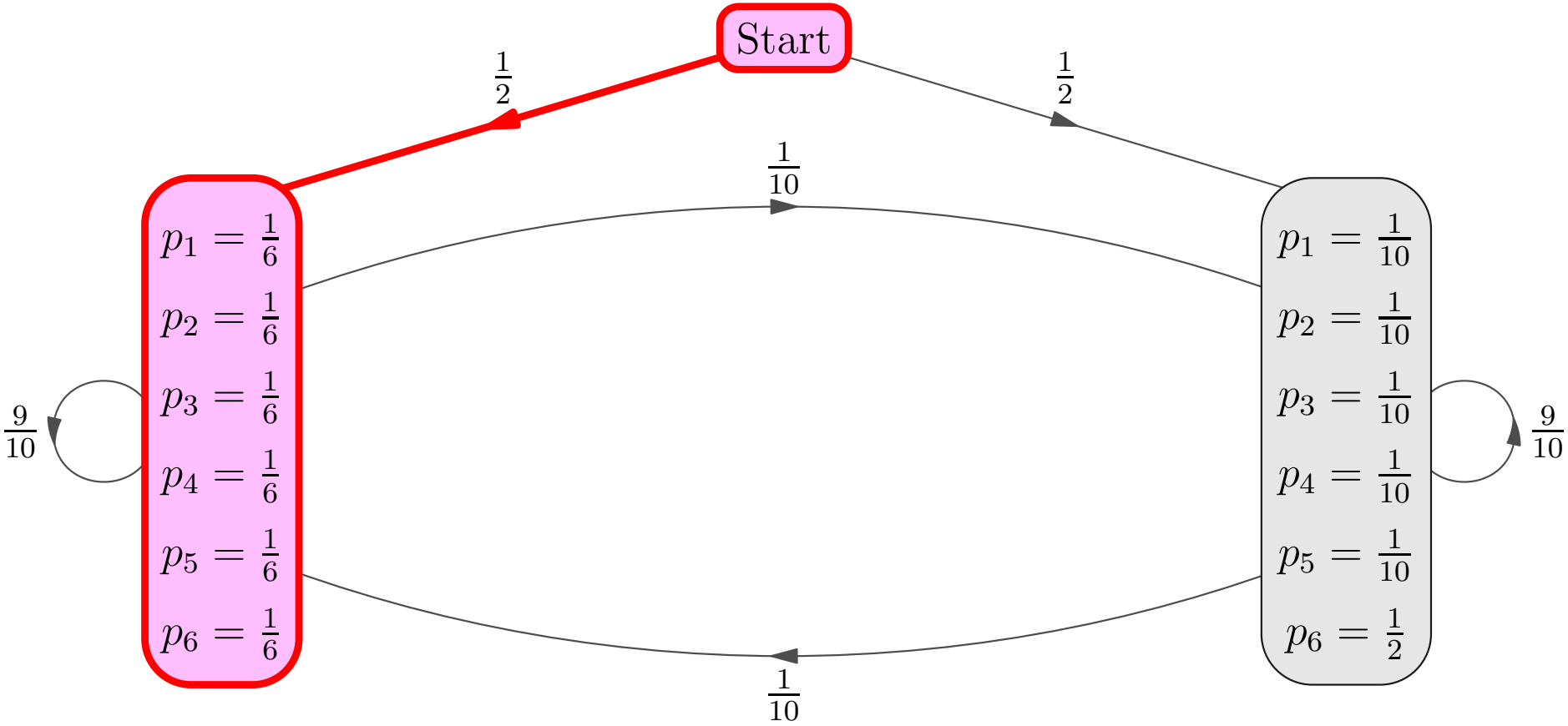
# Start State

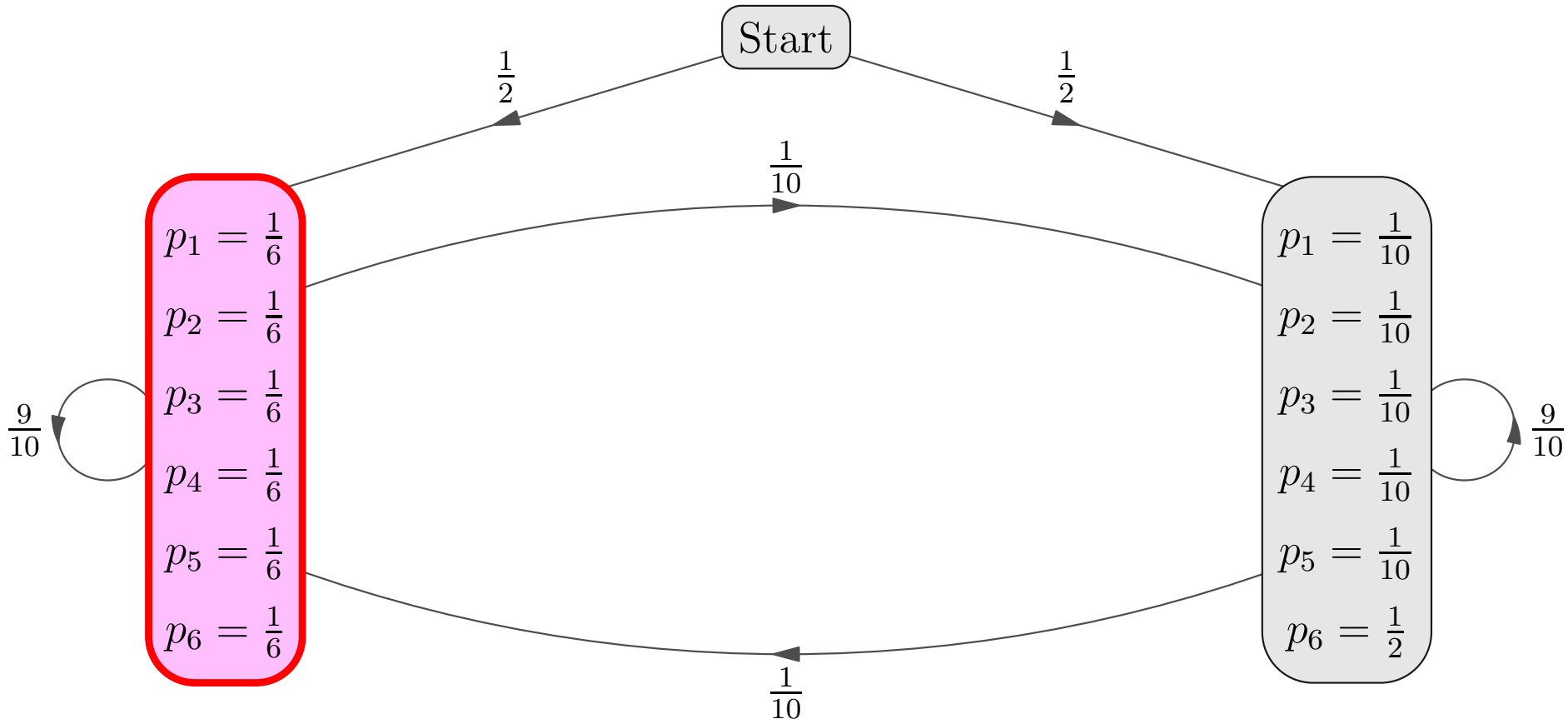

Start
$\frac{1}{2}$
$\frac{1}{2}$
$\frac{1}{10}$
$p_1 = \frac{1}{6}$
$p_2 = \frac{1}{6}$
$p_3 = \frac{1}{6}$
$p_4 = \frac{1}{6}$
$p_5 = \frac{1}{6}$
$p_6 = \frac{1}{6}$
$p_1 = \frac{1}{10}$
$p_2 = \frac{1}{10}$
$p_3 = \frac{1}{10}$
$p_4 = \frac{1}{10}$
$p_5 = \frac{1}{10}$
$p_6 = \frac{1}{2}$
$\frac{9}{10}$
$\frac{9}{10}$
$\frac{1}{10}$

# Start State



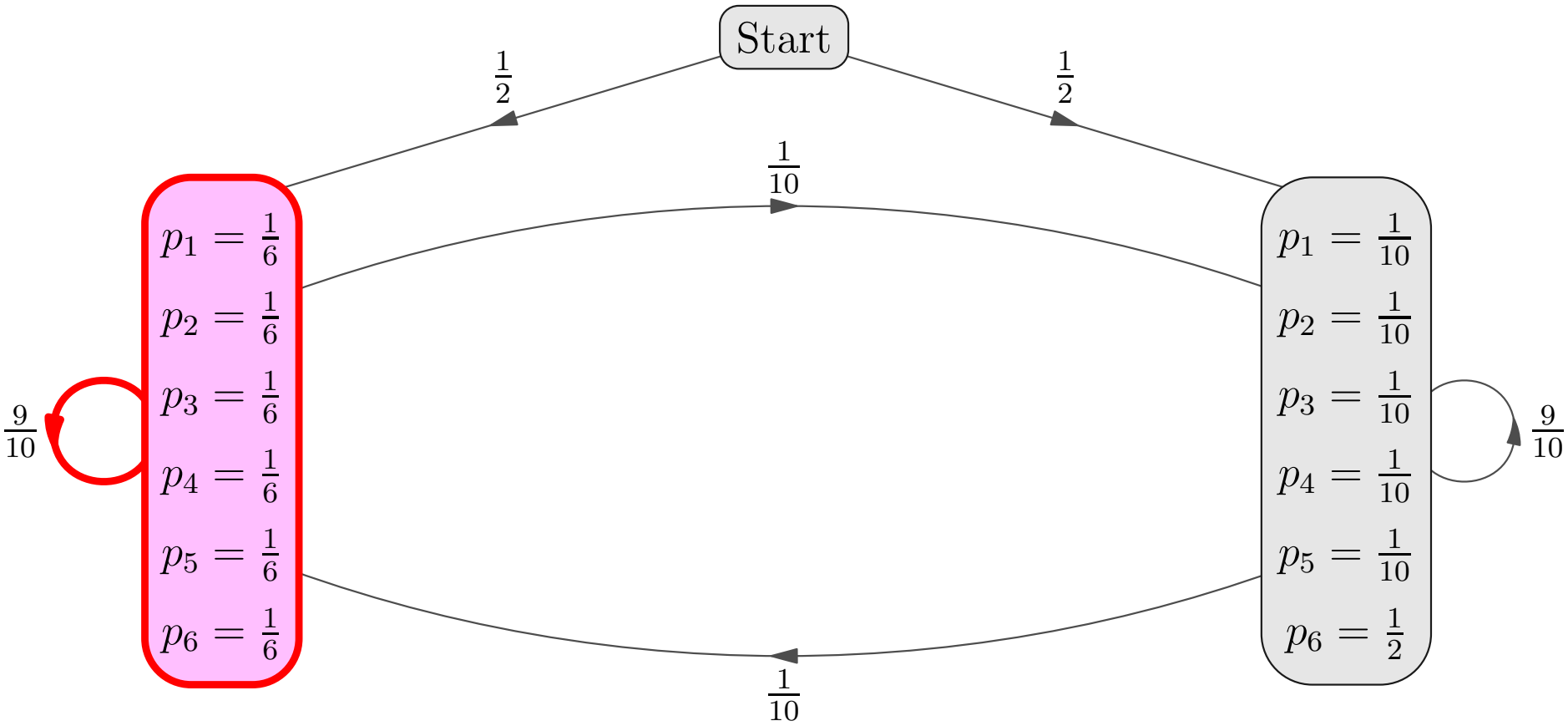

H

# Start State




4
H

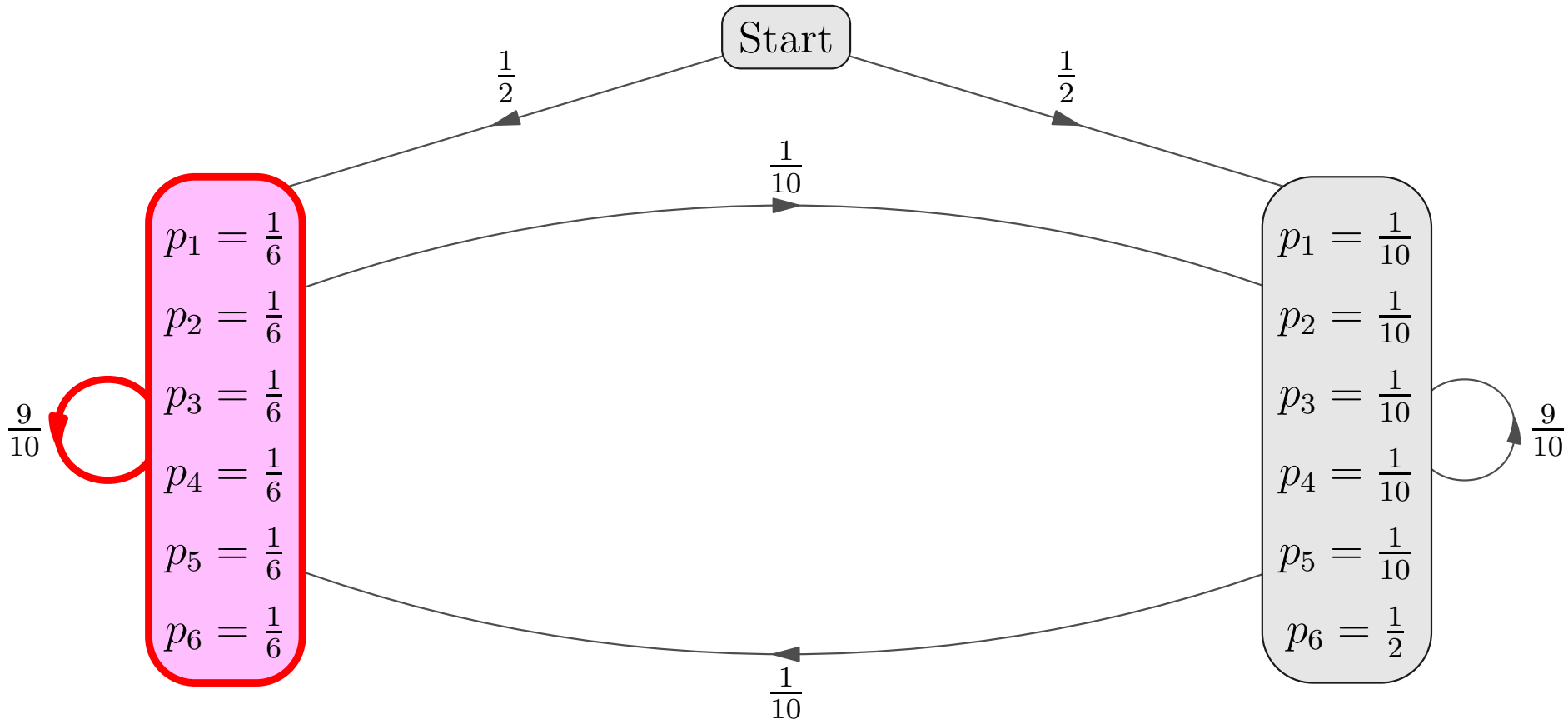
# Start State




4

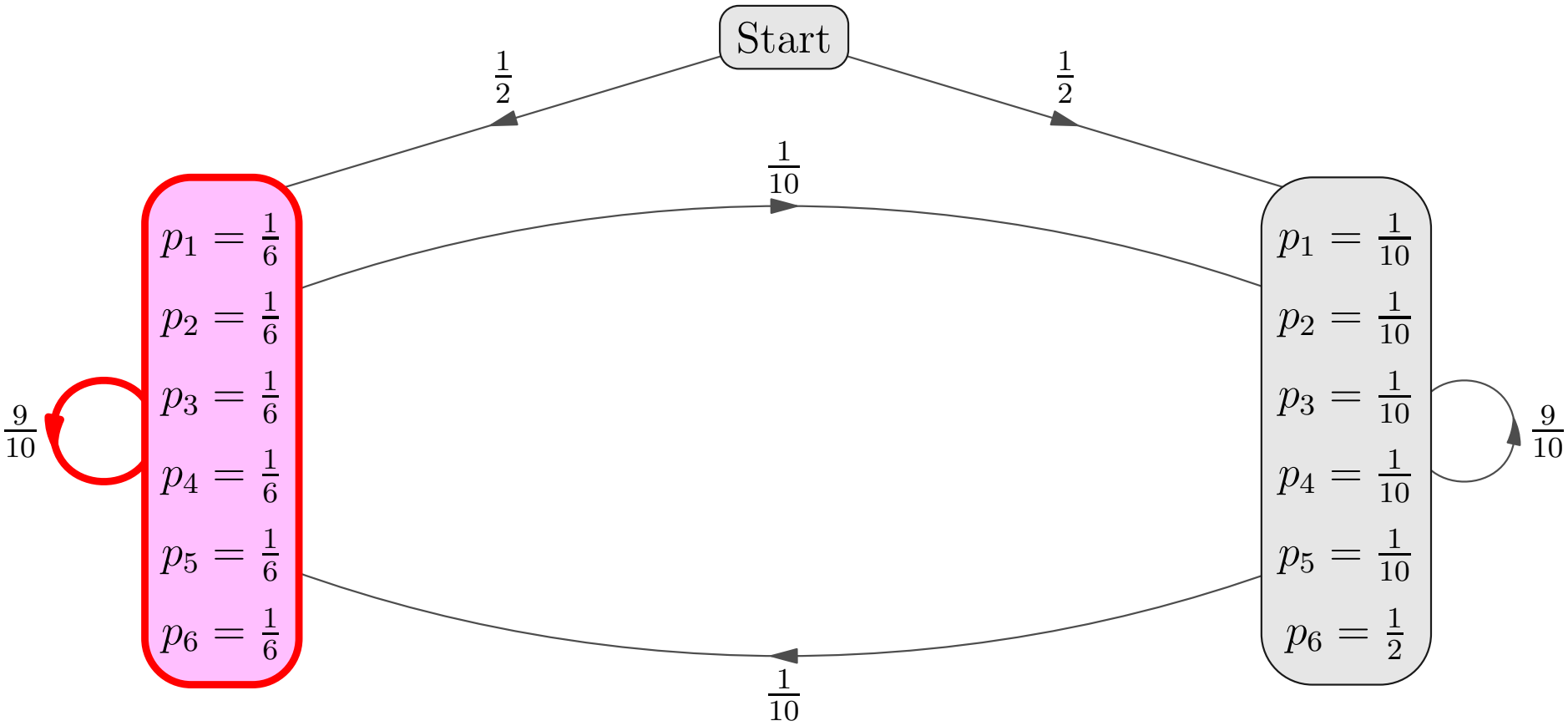
HH

# Start State



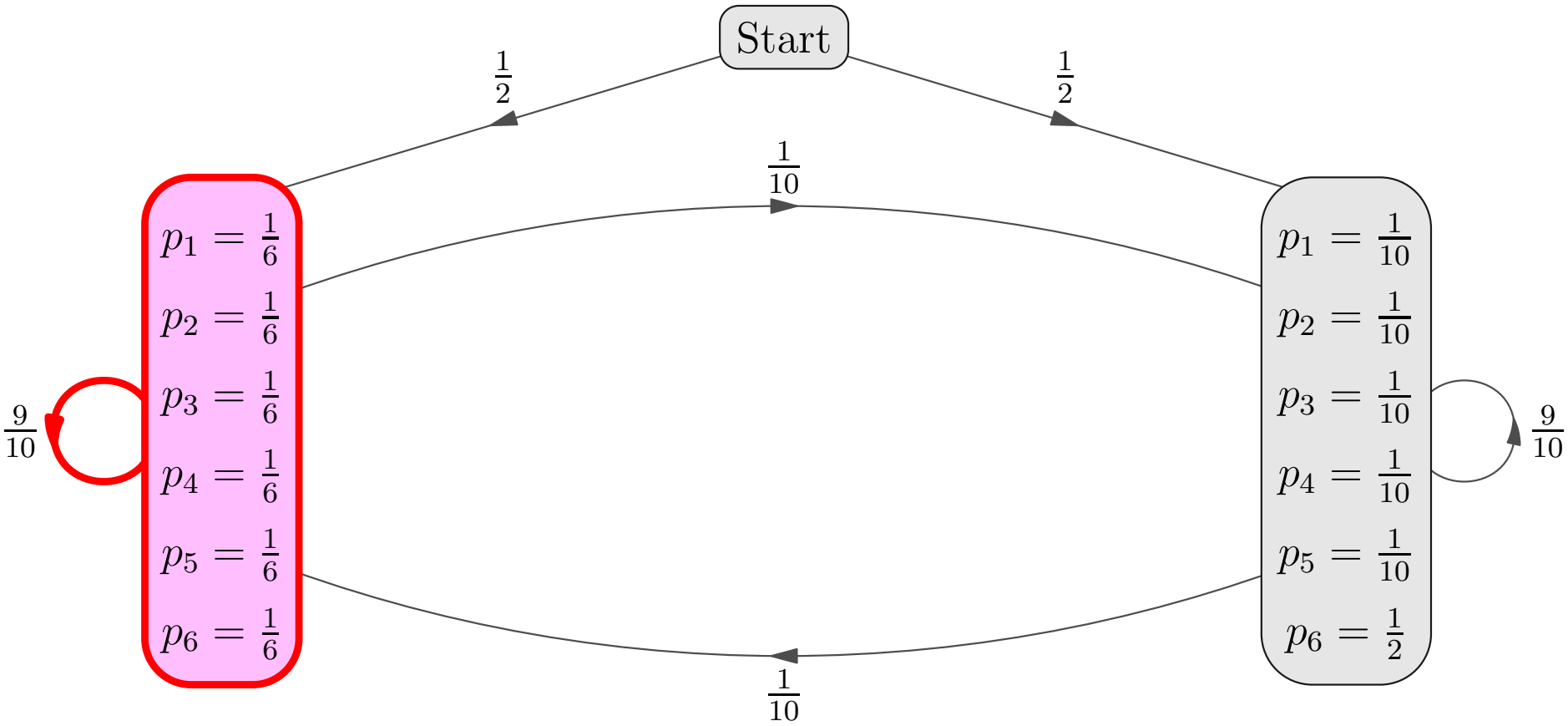

41
HH

# Start State



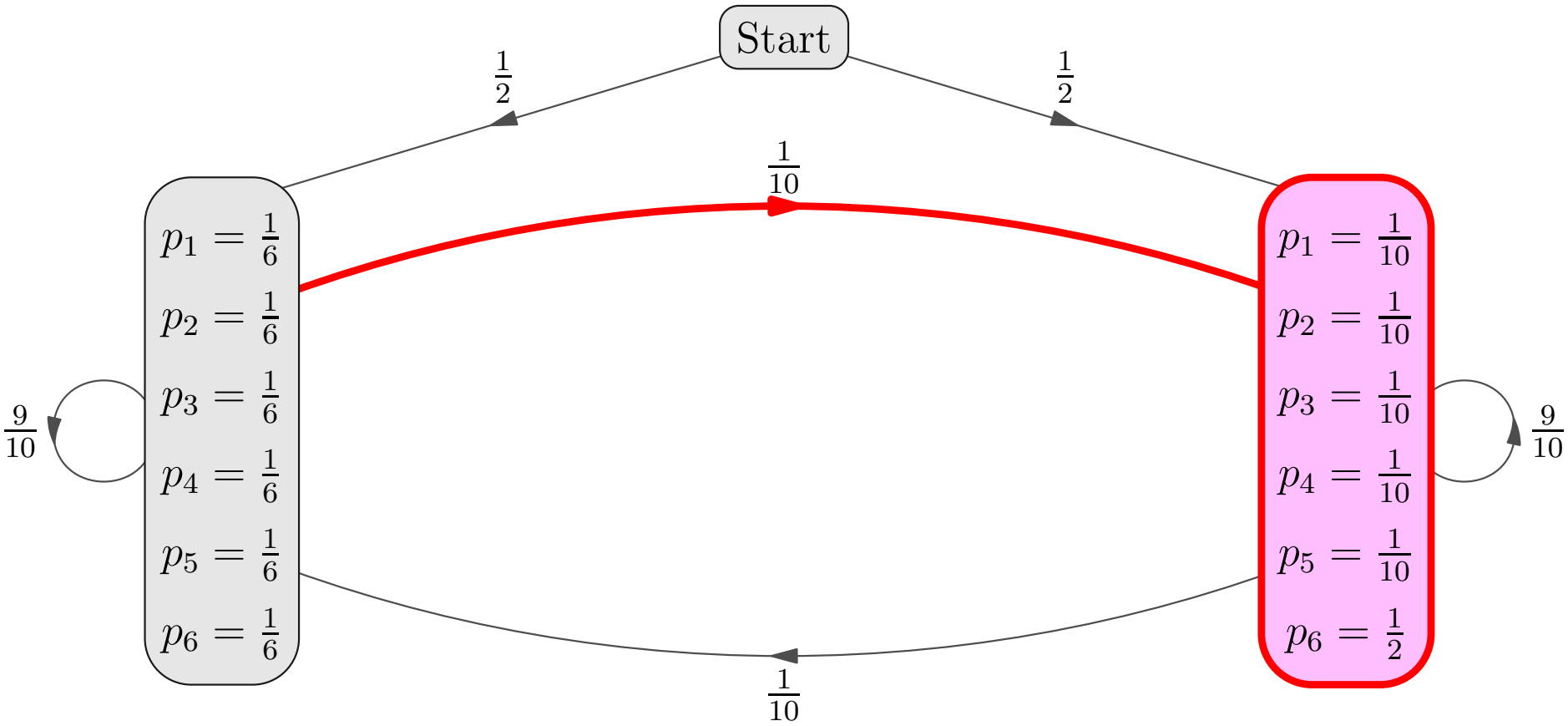

41
HHH

# Start State



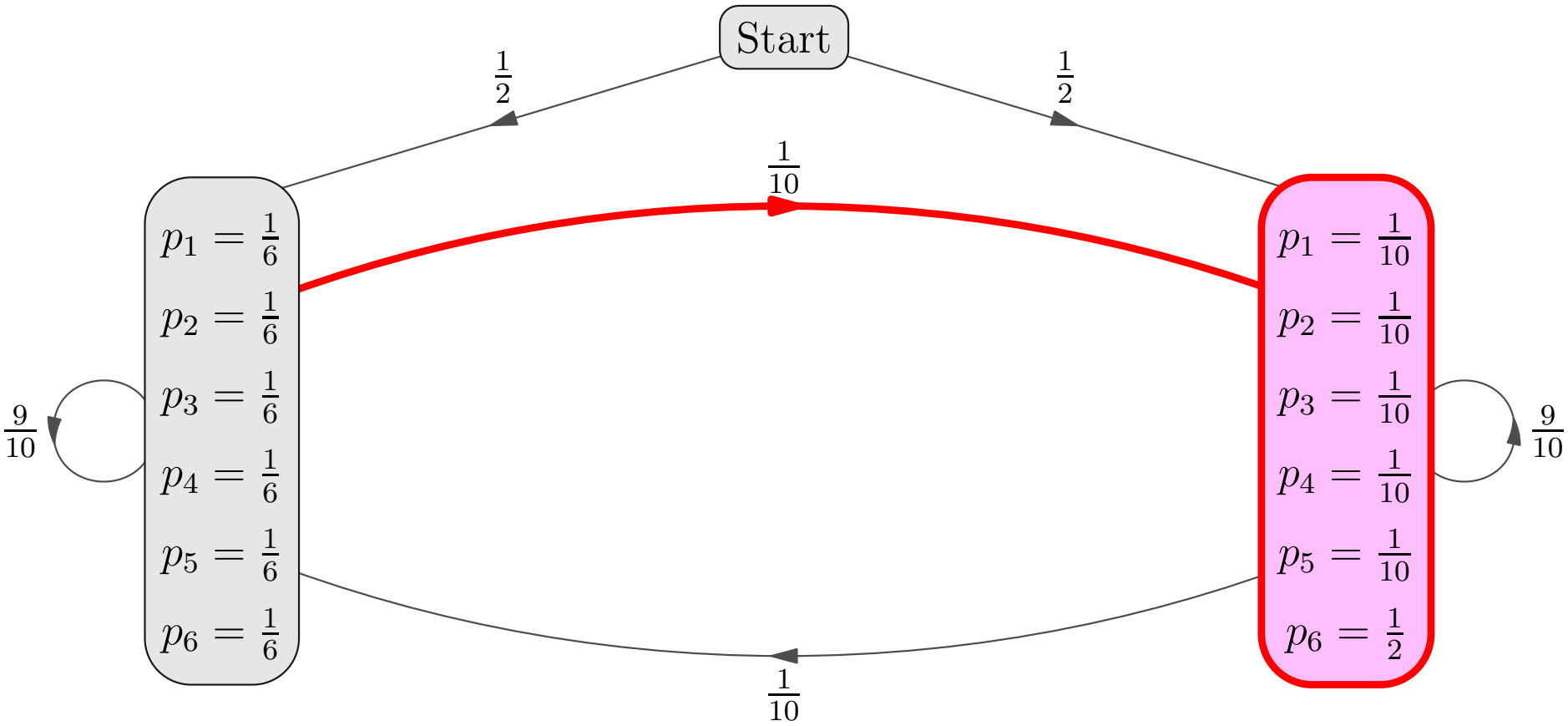

415
HHH

# Start State



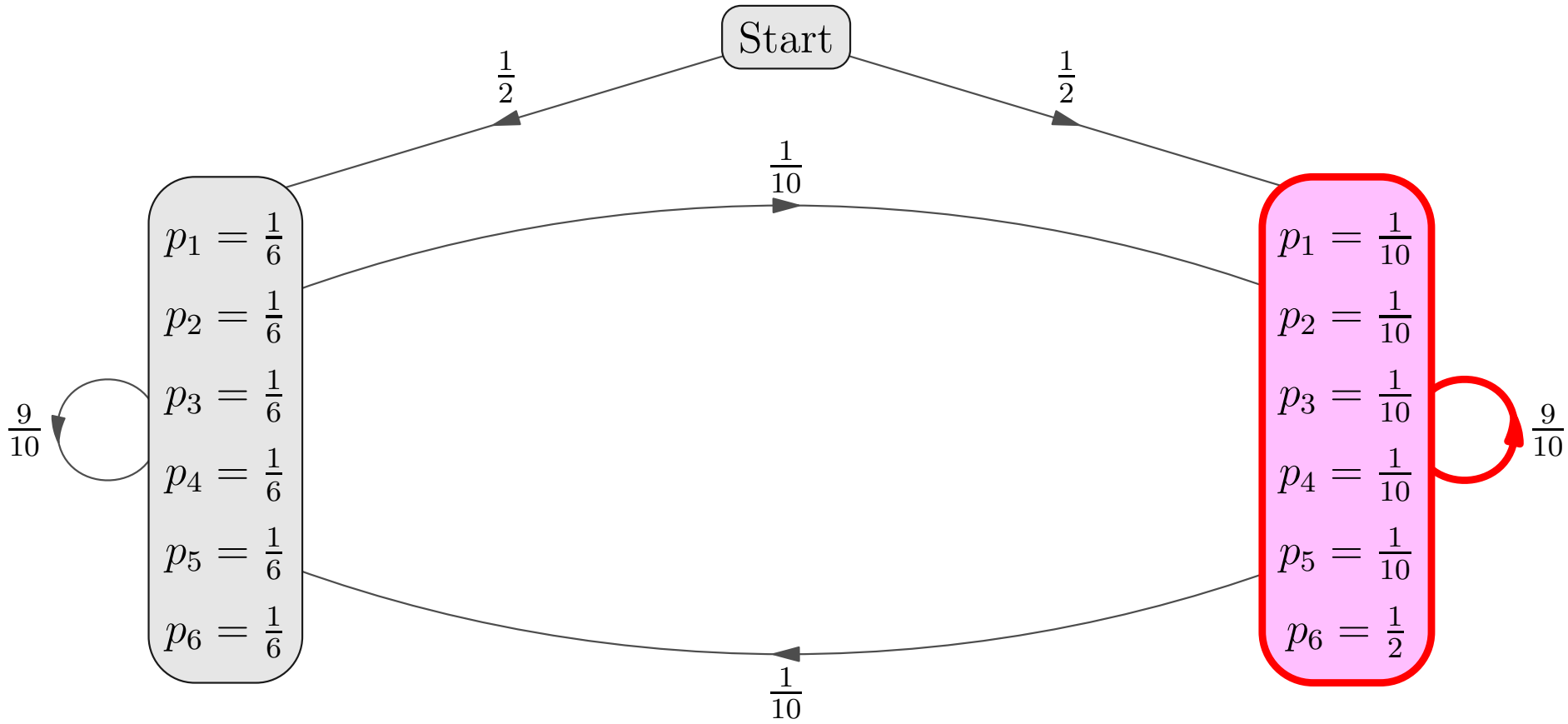

415
HHHD

# Start State



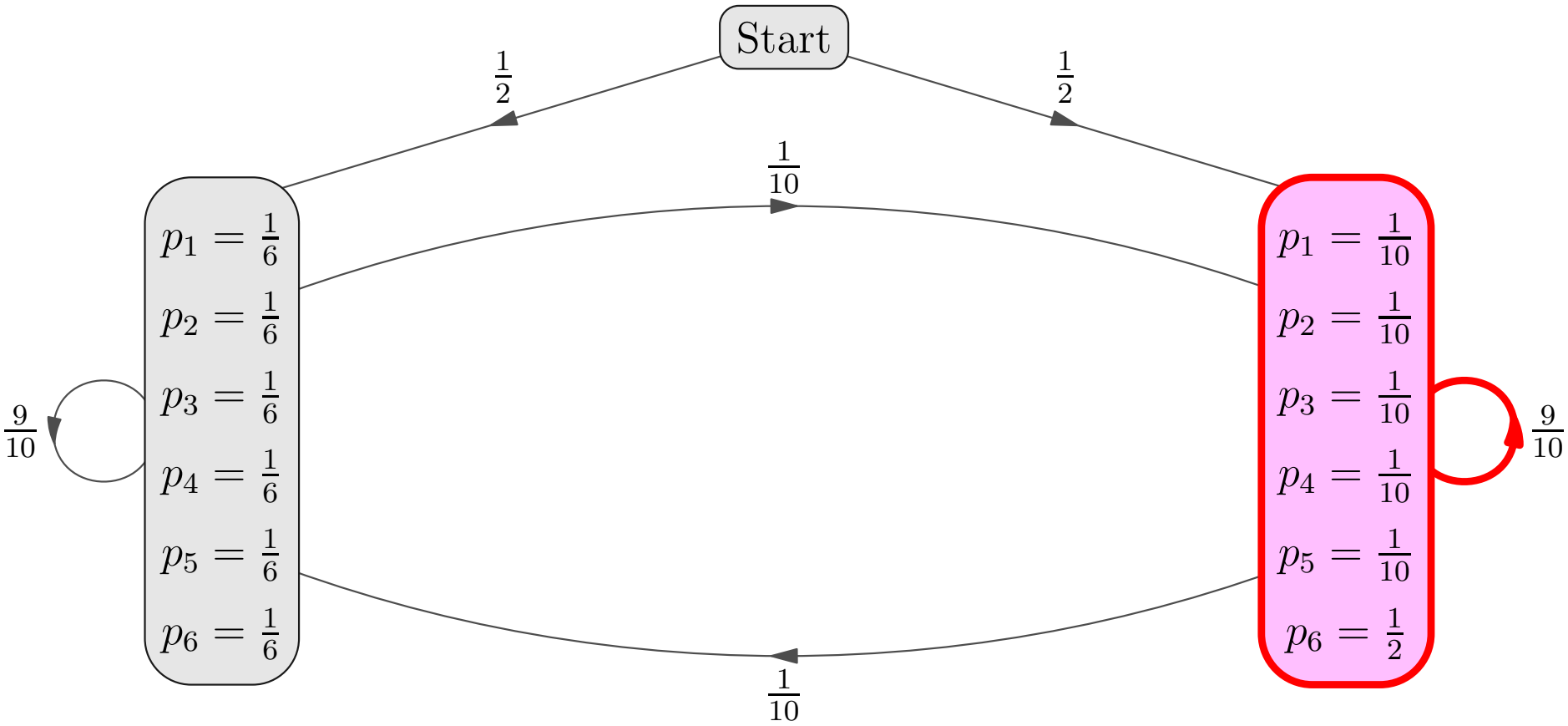

4156
HHHD

# Start State




41566
HHHDD

# Start State




415663
HHHDDD

# Parameter of HMM

- The **emission probability** is the probability of emitting a letter $\zeta$ in state $i$ and is denoted

$$e_i(\zeta) = \mathbb{P}\left(\zeta | q_t = i\right)$$

- The **transition probability** of moving from state $i$ to state $j$ is denoted

$$a_{ij} = \mathbb{P}\left(q_t = j | q_{t-1} = i\right)$$

- The set of emission and transition probabilities we denote by $\boldsymbol{\theta}$

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# Viterbi Algorithm

- To find the most likely set of states associated with a sequence $\boldsymbol{\xi}$ we can use a classic dynamic programming algorithm
- This is known as the Viterbi algorithm
- Let $v_i(t)$ be the probability of being in state $i$ at time $t$
- Initially $v_0(0) = 1$ and $v_i(0) = 0$ where state 0 is the initial state
- We iteratively compute for each state $i \in \mathcal{Q}$

$$v_i(t) = e_i(\xi_t) \max_j v_j(t-1)\, a_{ji}$$

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# Most Likely Path

- To compute the optimal path it helps to maintain a pointer

$$\mathrm{ptr}_t(i) = \operatorname*{argmax}_j \, v_j(t-1)\, a_{ji}$$

- The termination condition for Viterbi is

$$\mathbb{P}\left(\boldsymbol{\xi}, \boldsymbol{q}^*\right) = \max_i v_i(T)$$

$$q_T^* = \operatorname*{argmax}_i v_i(T)$$

- The backward algorithm is

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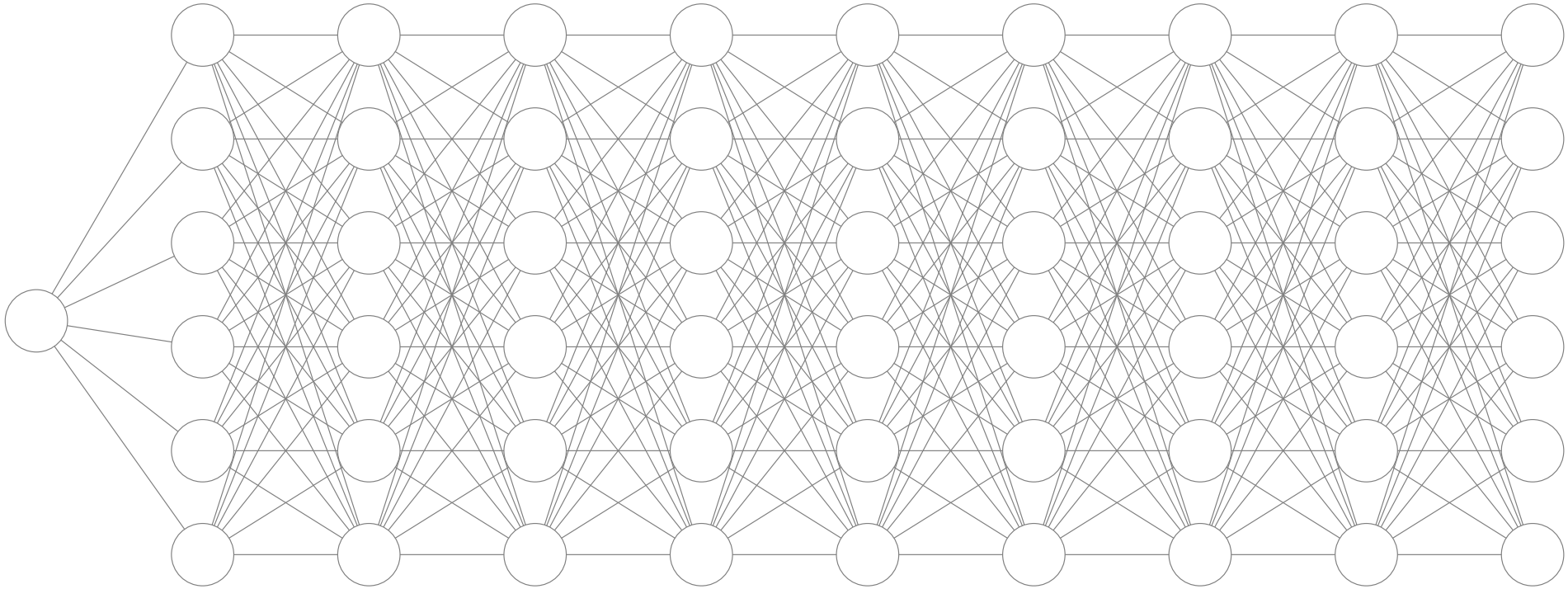
$$v_i(t) = e_i(\xi_t)\ \max_j v_j(t-1)\, a_{ji}, \quad v_0(0) = 1$$



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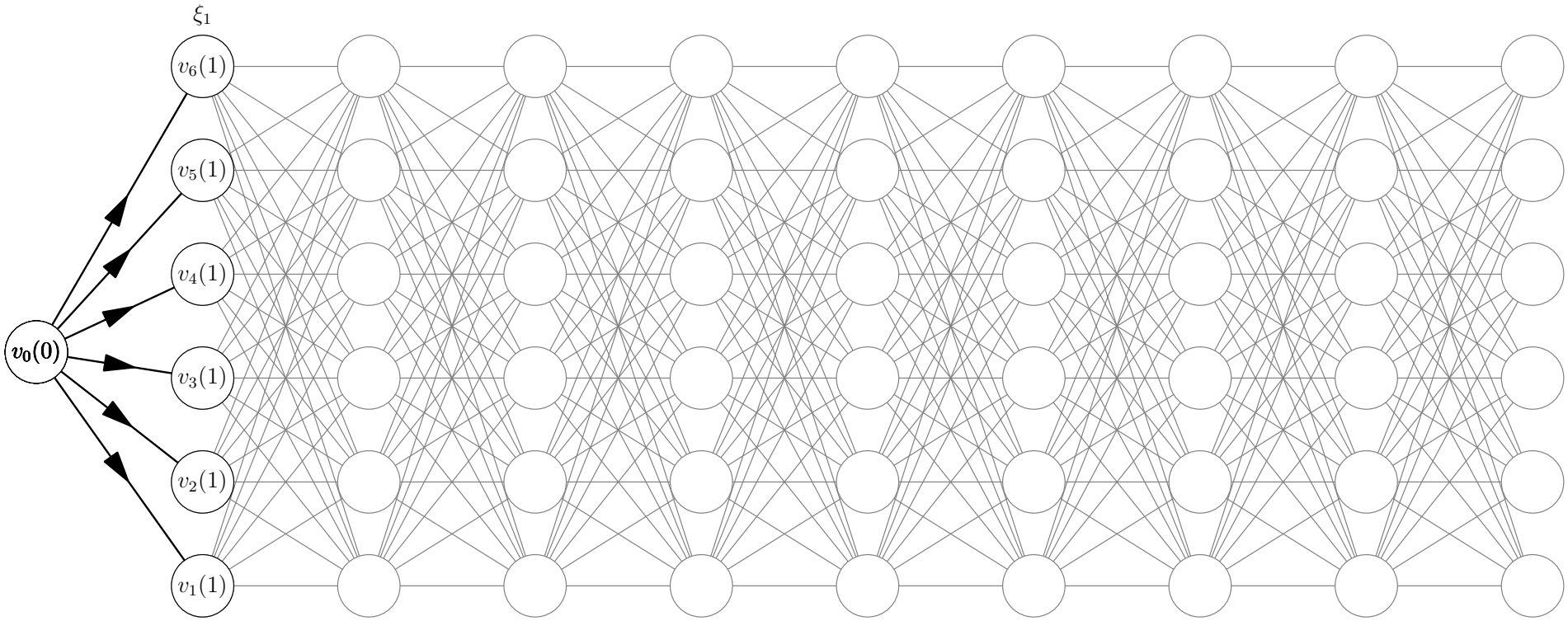

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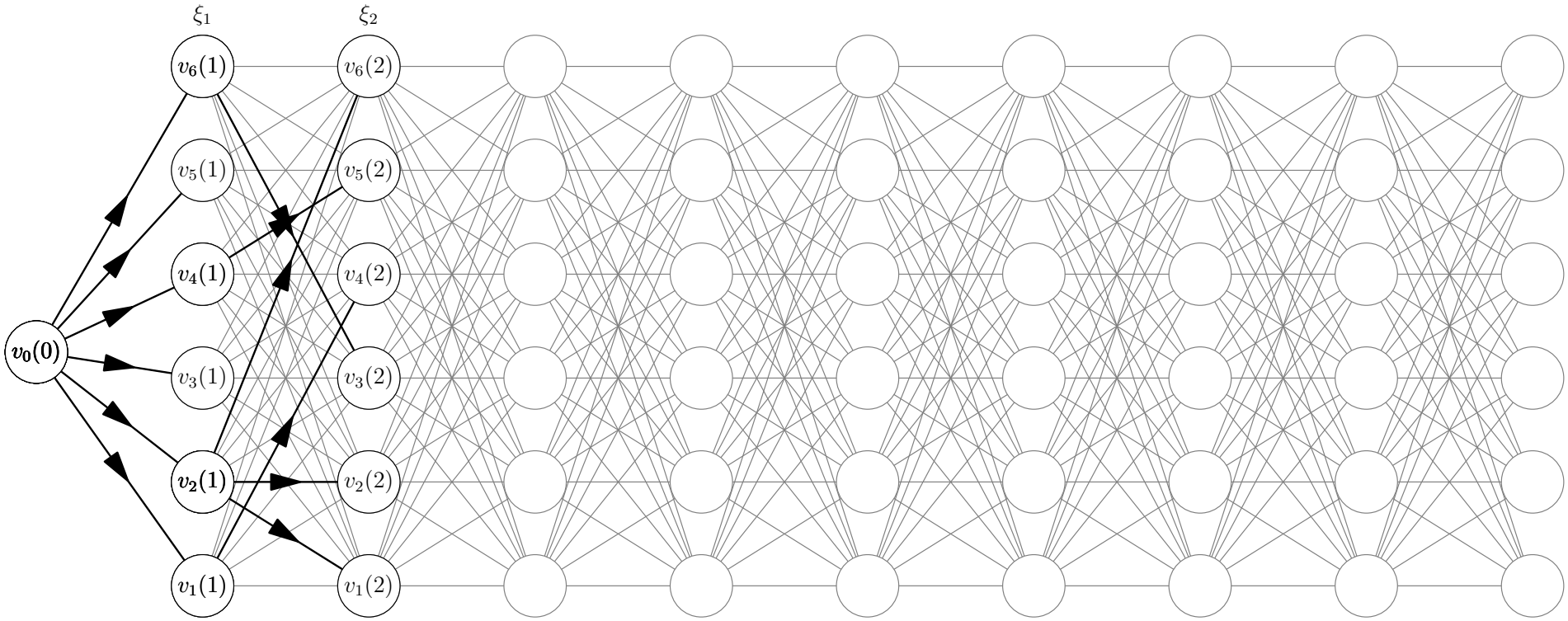

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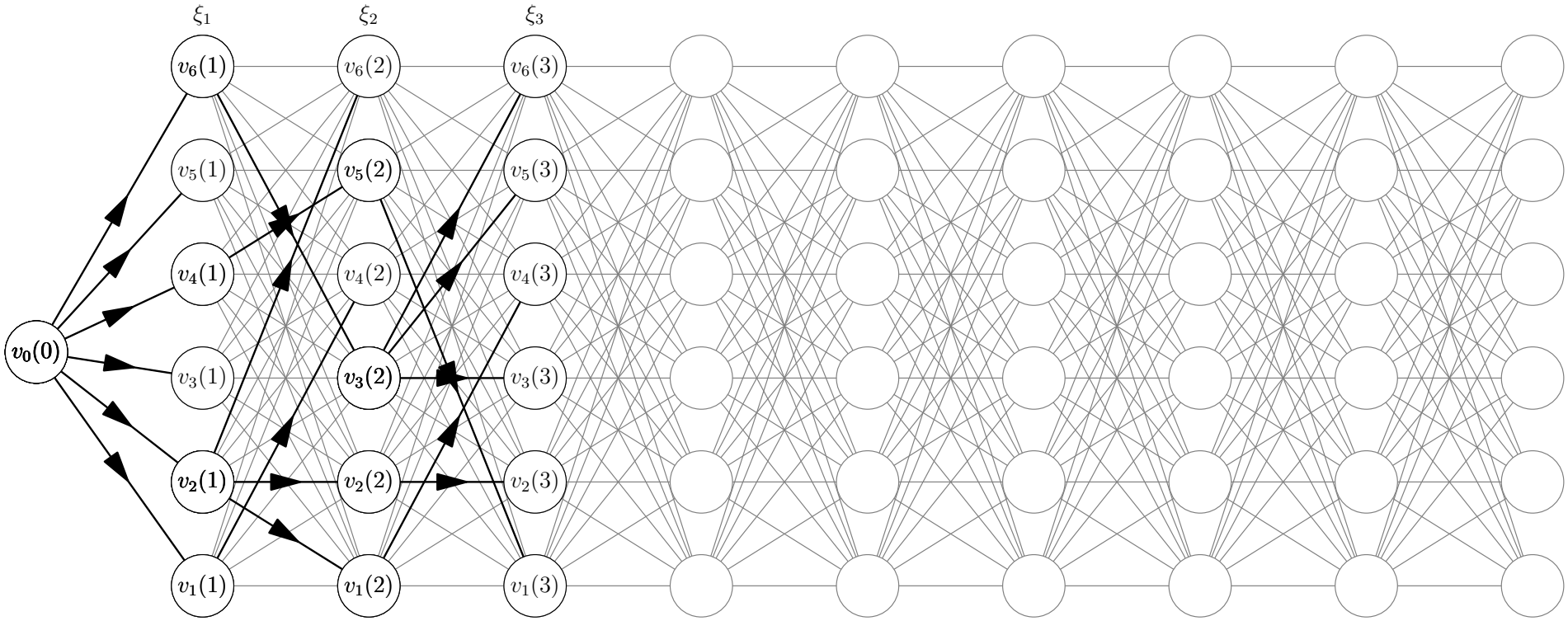

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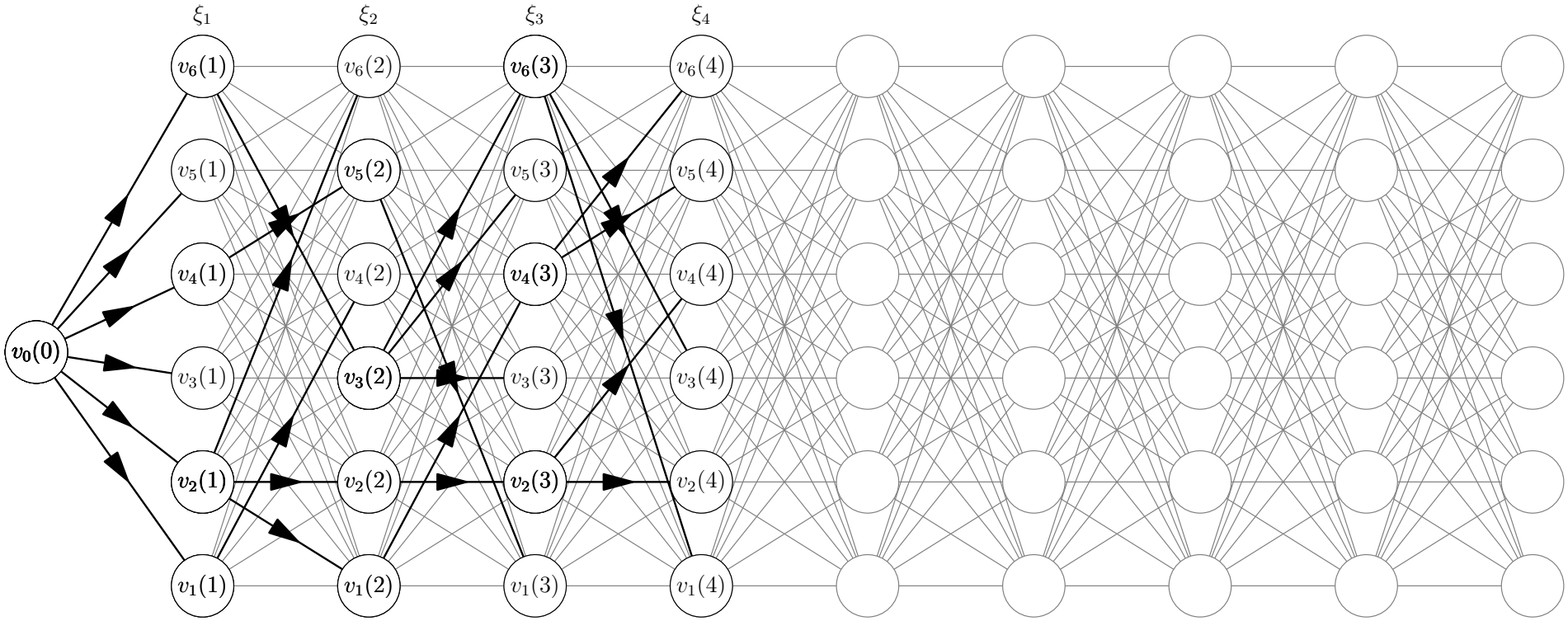

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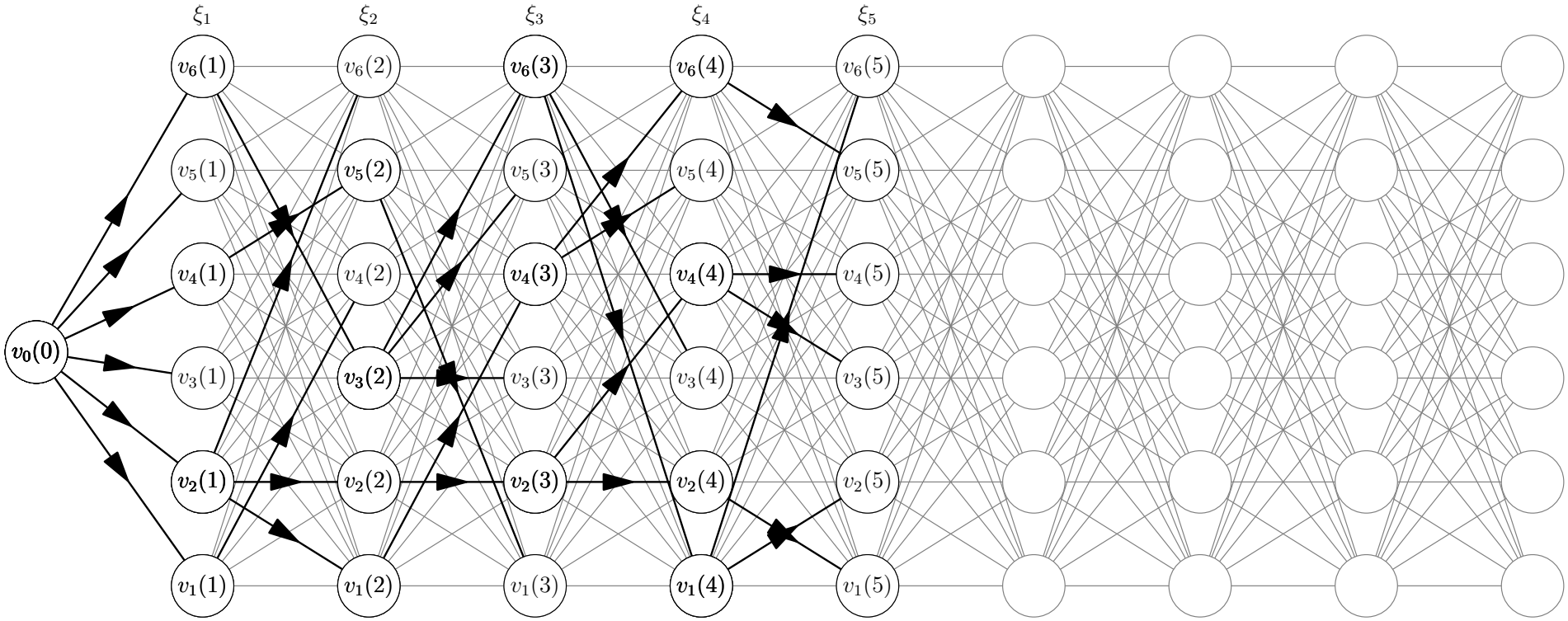

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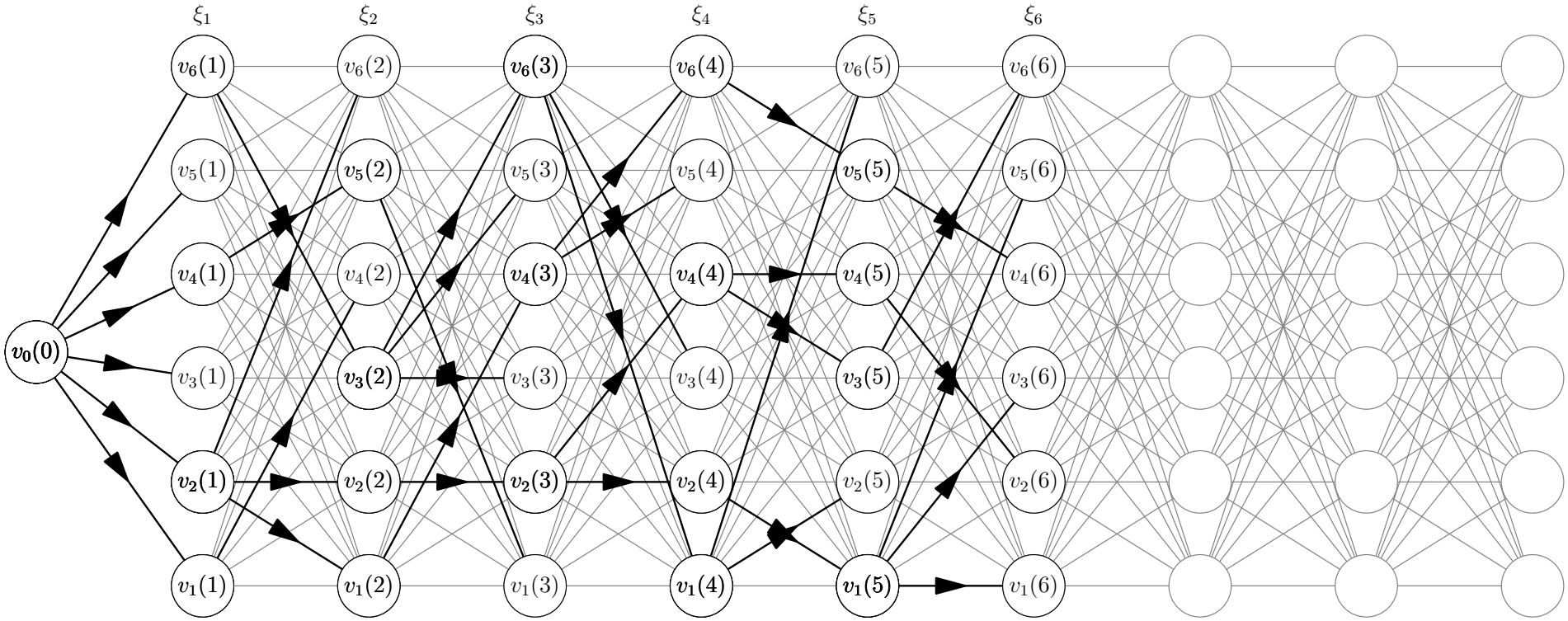

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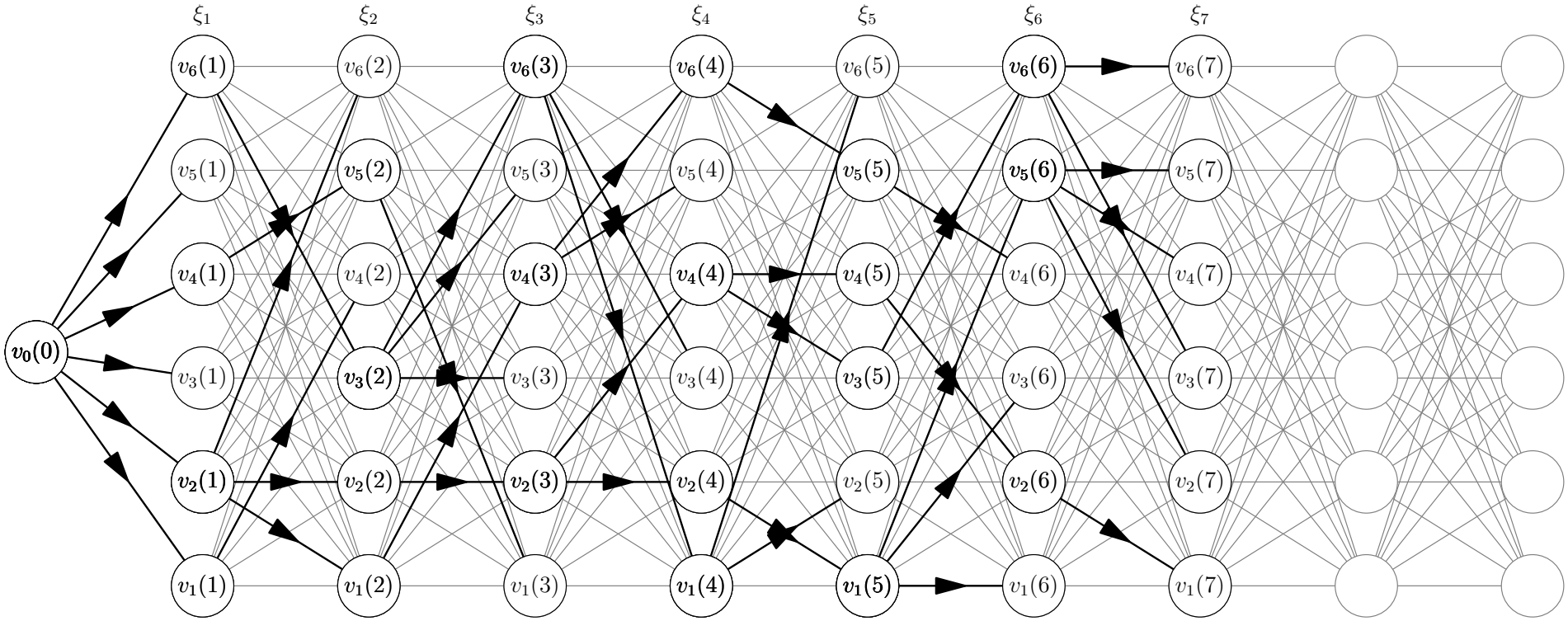

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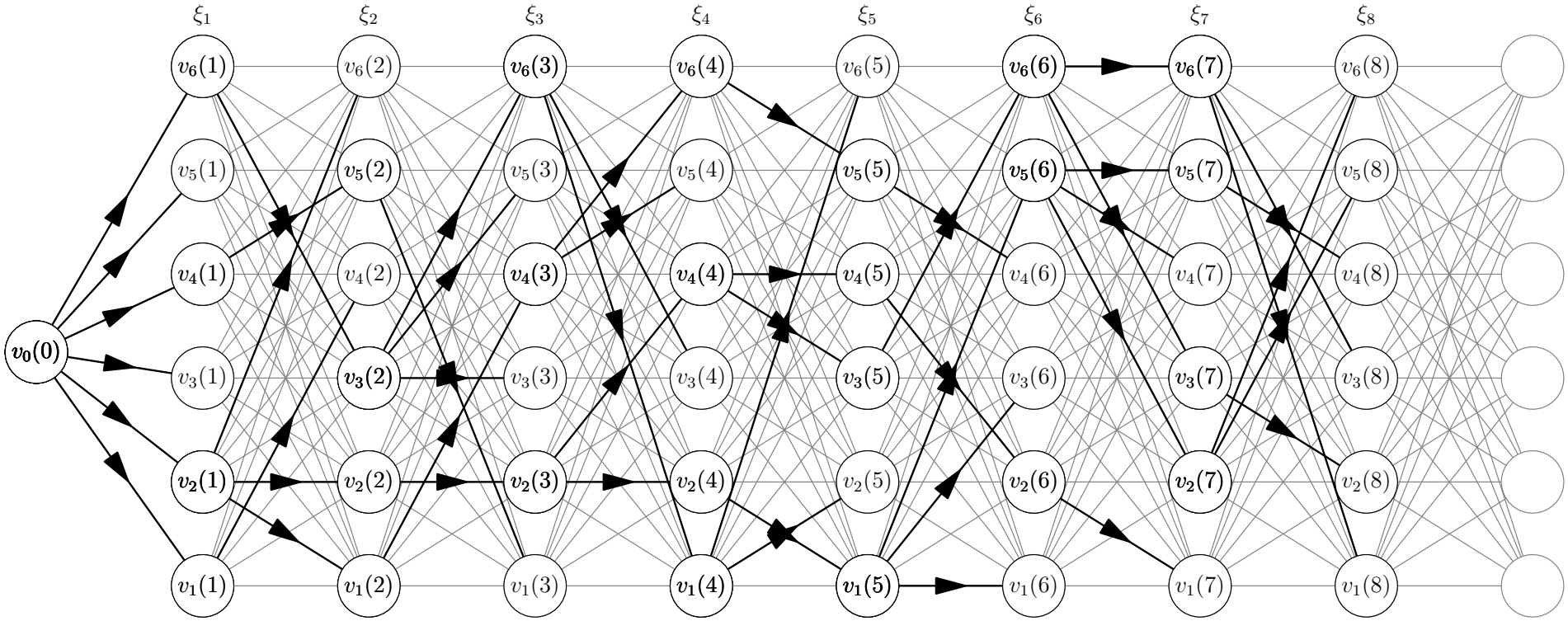

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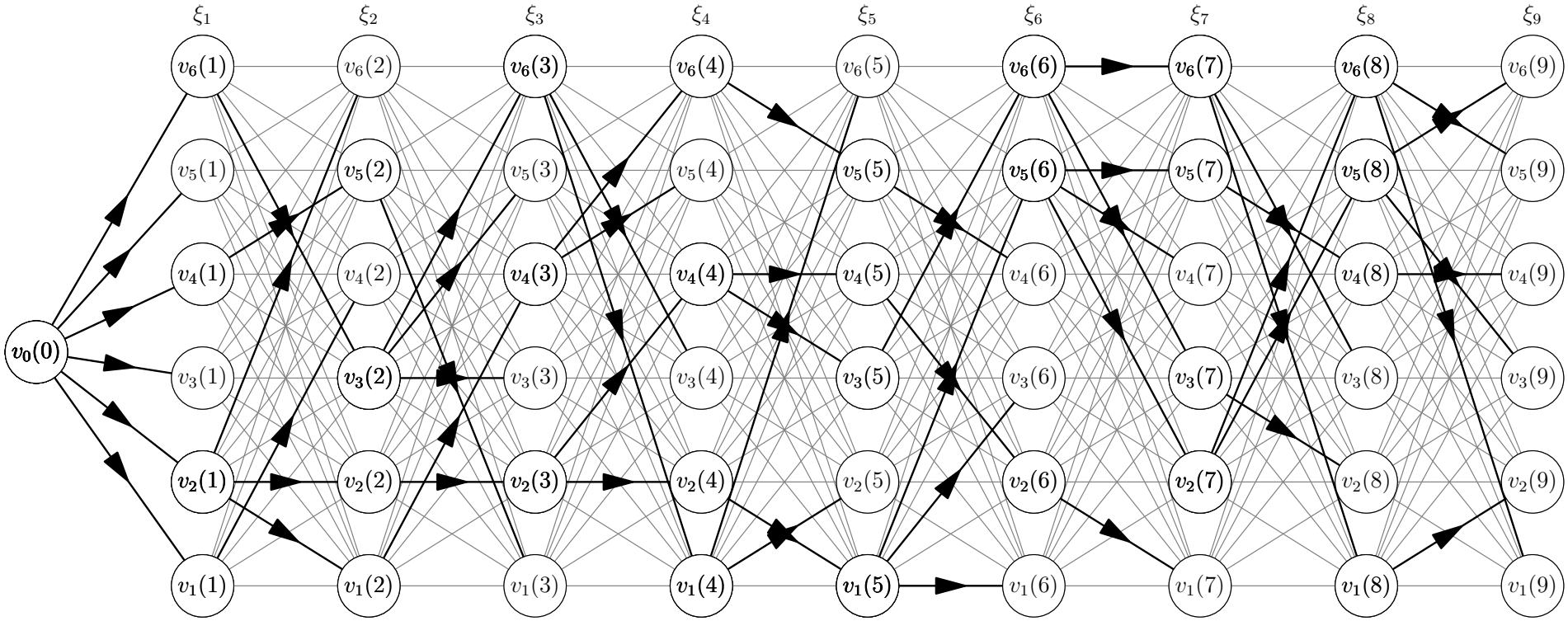

# Using Log-Probabilities

- Numerically products of probabilities underflow: if $P = \prod_{i=1}^{N} p_i$ with $p_i = 0.1$ and $n = 400$ then $P = (0.1)^{400}$

- Now if $P = \prod_{i=1}^{N} p_i$ then $\log(P) = \sum_{i=1}^{N} \log(p_i)$ so with $p_i = 0.1$ and $n = 400$ then $\log(P) = 400 \times \log(0.1)$

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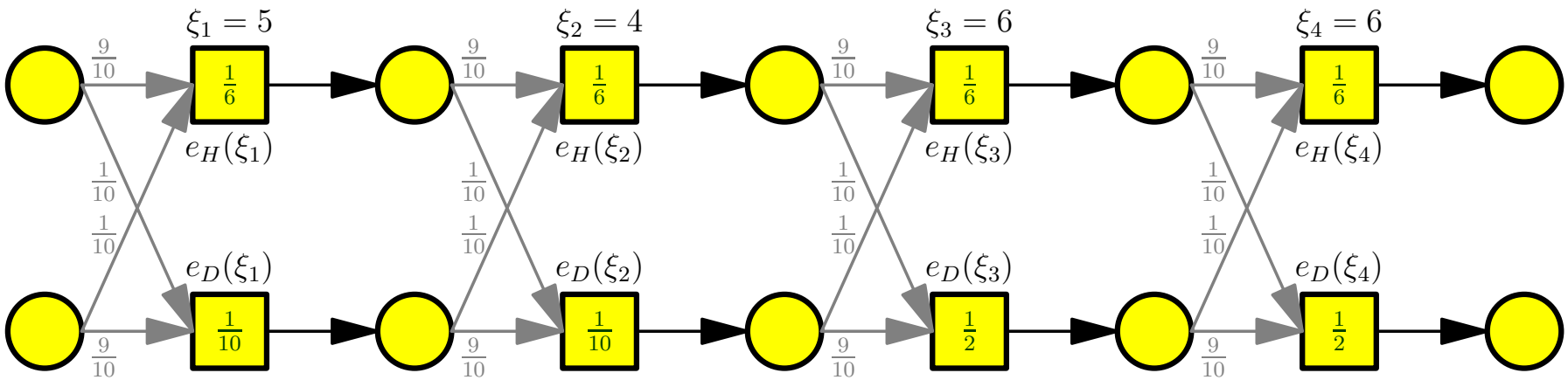

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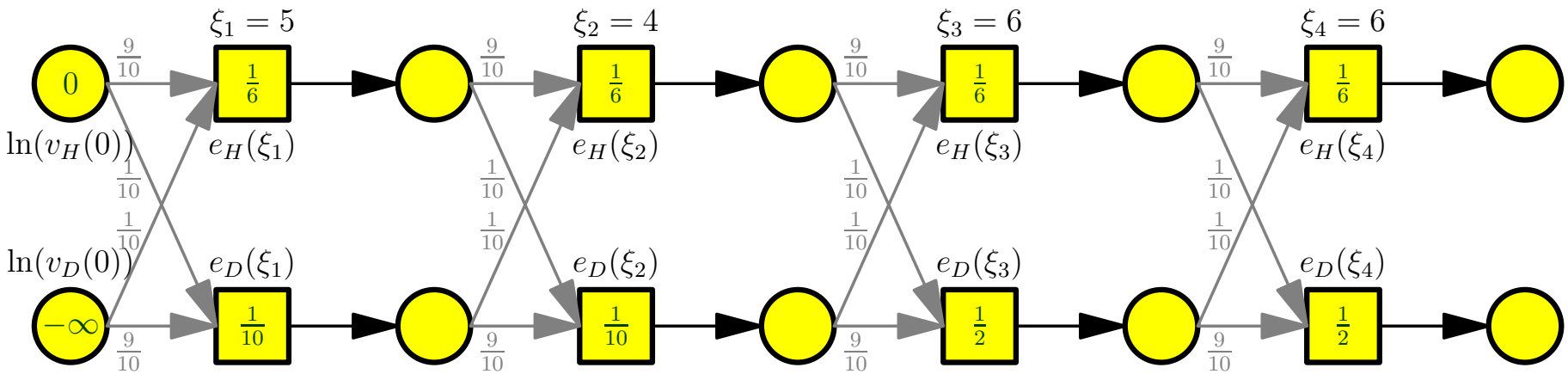

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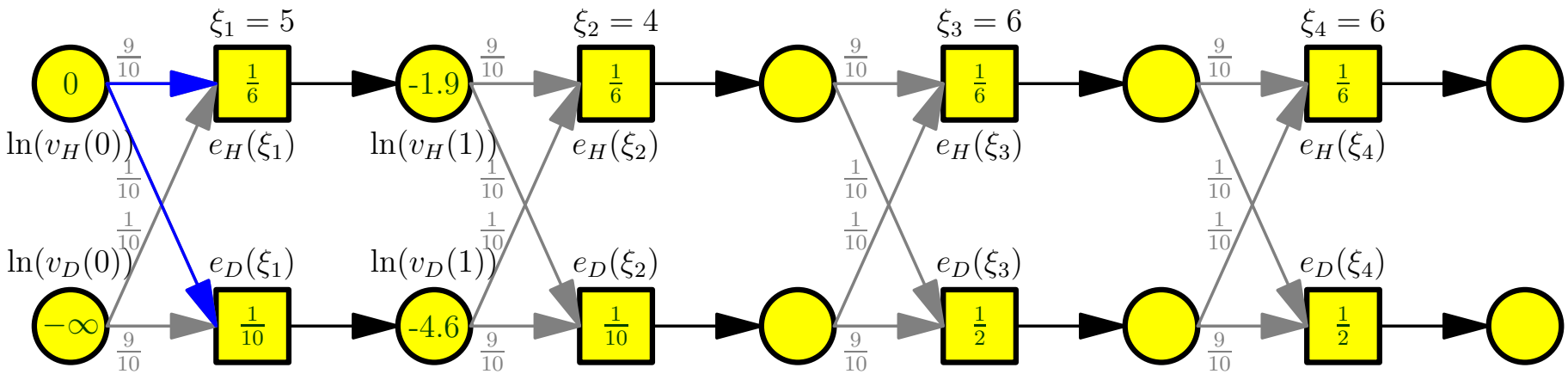

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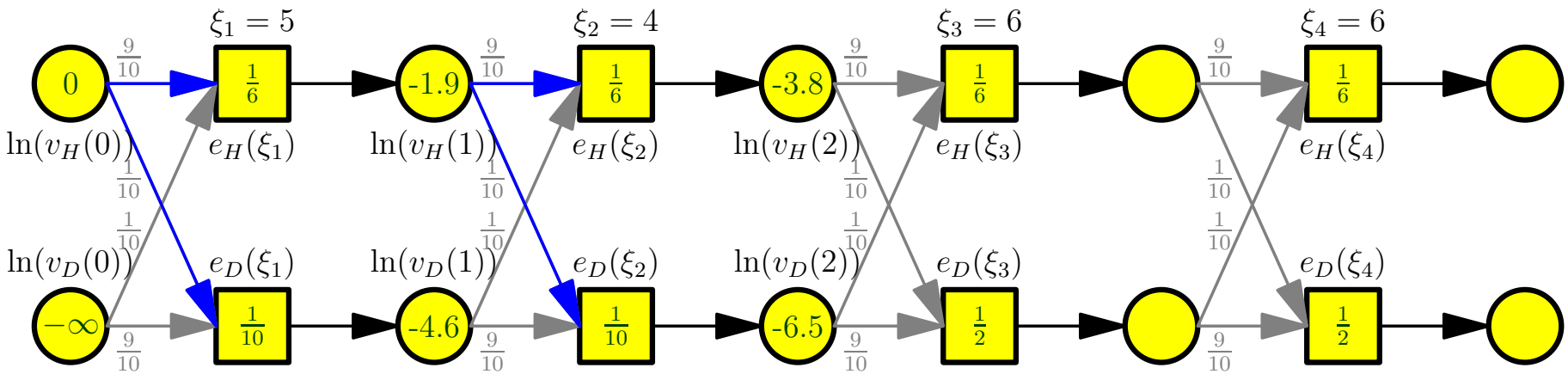

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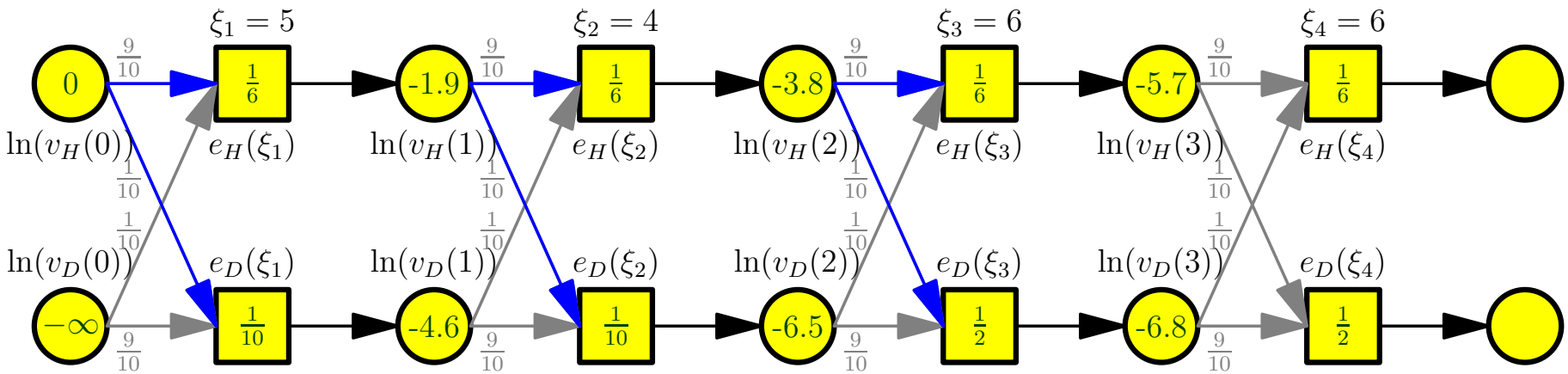

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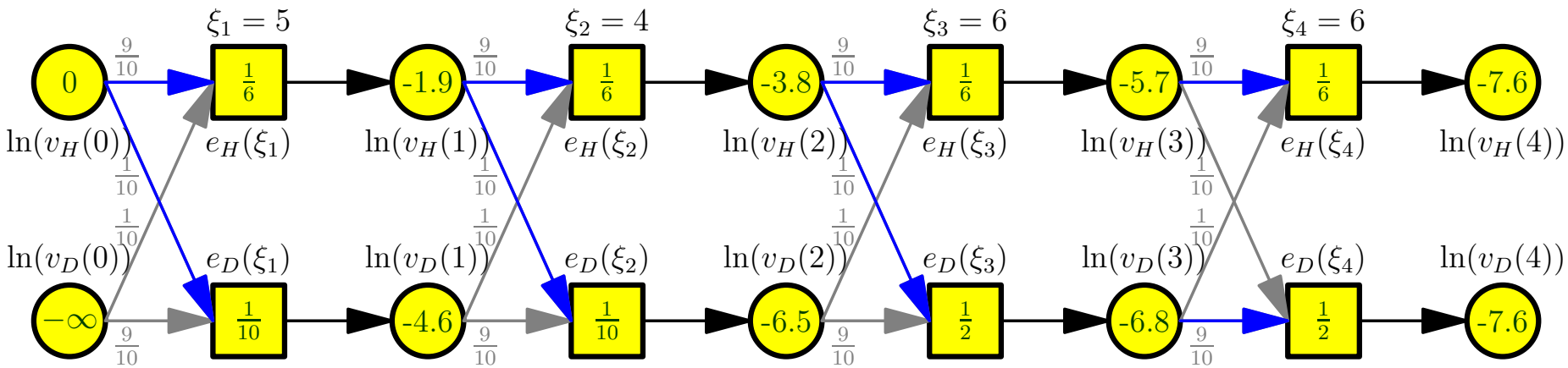

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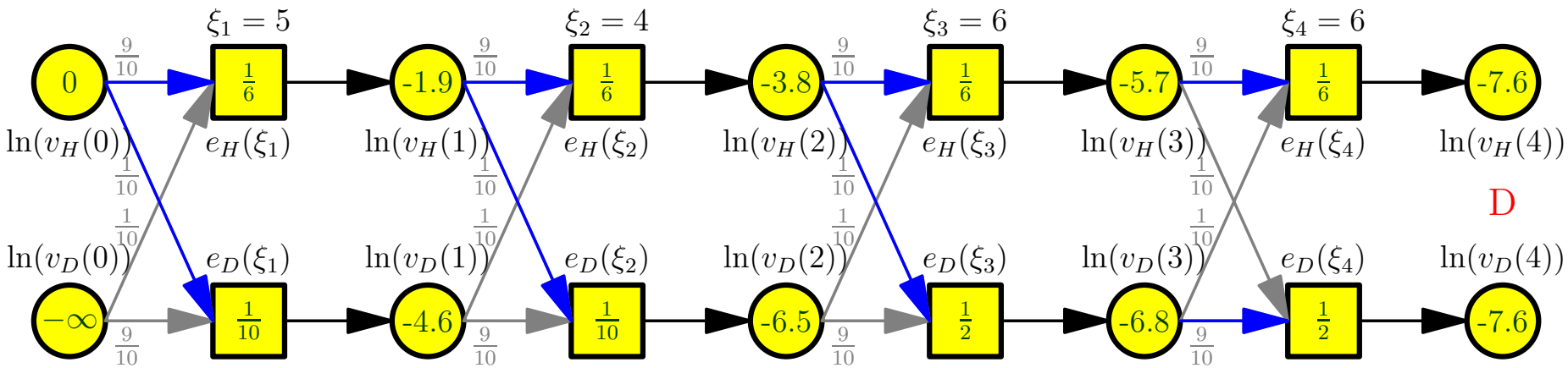

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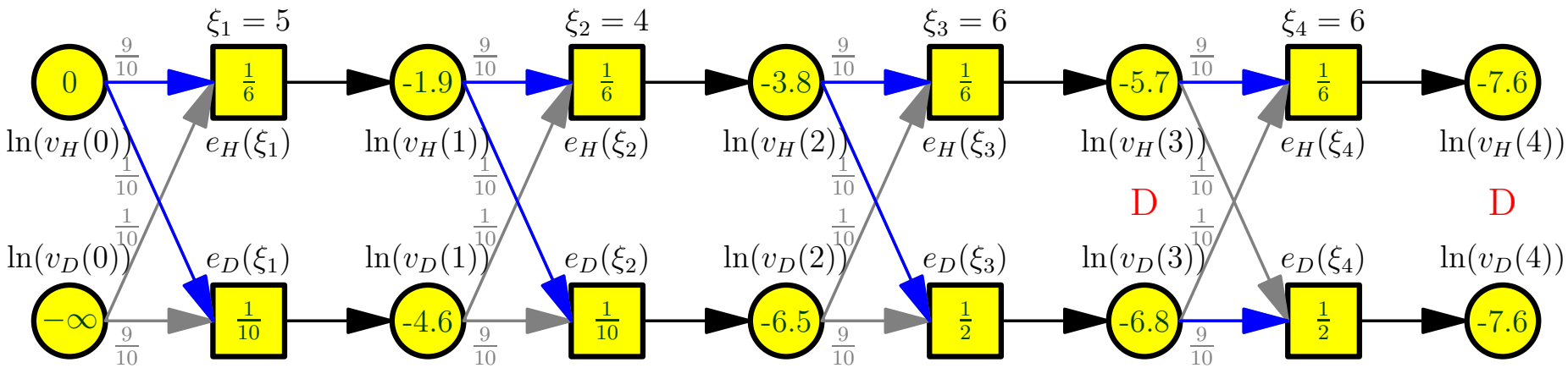

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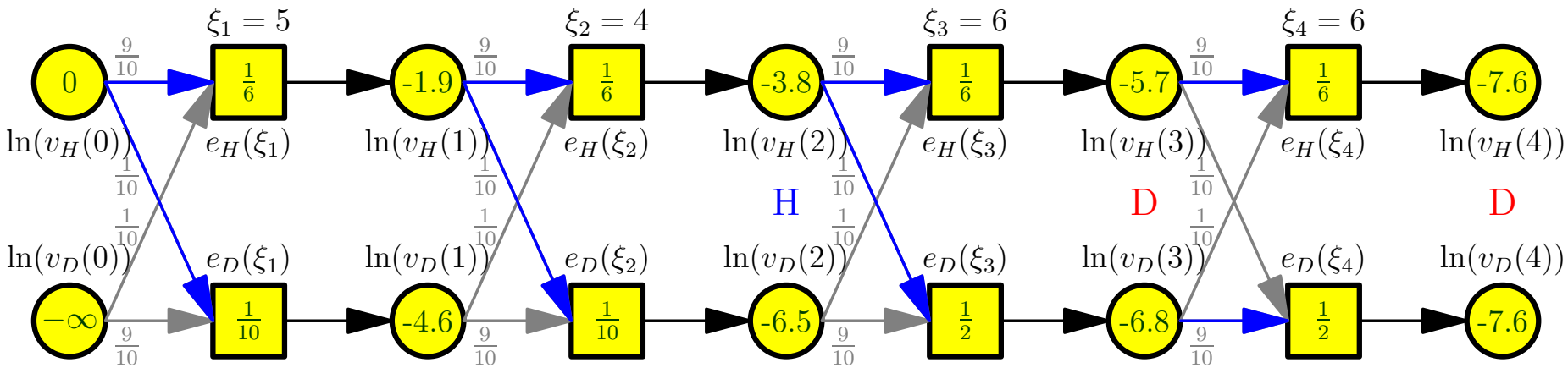

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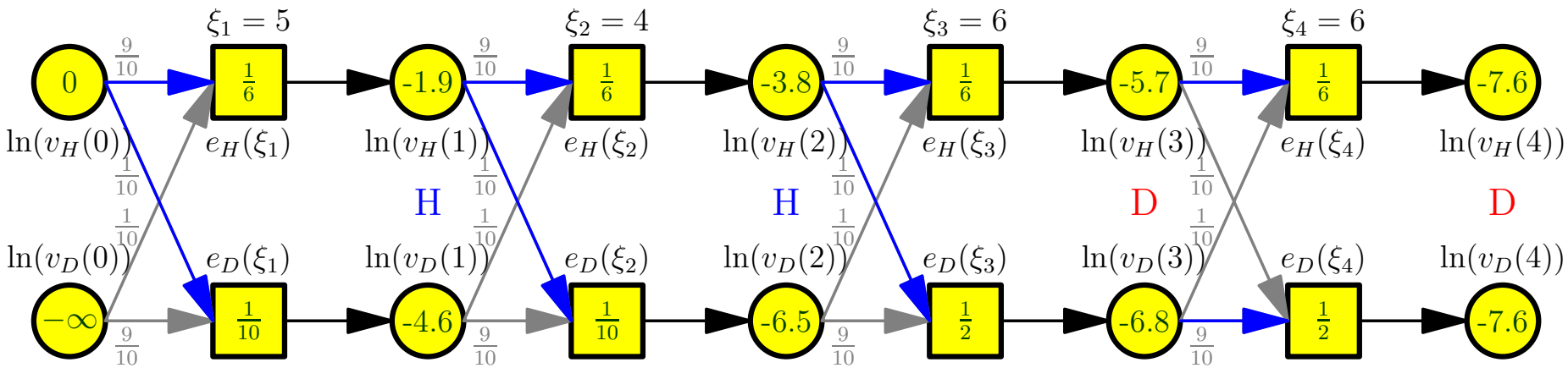

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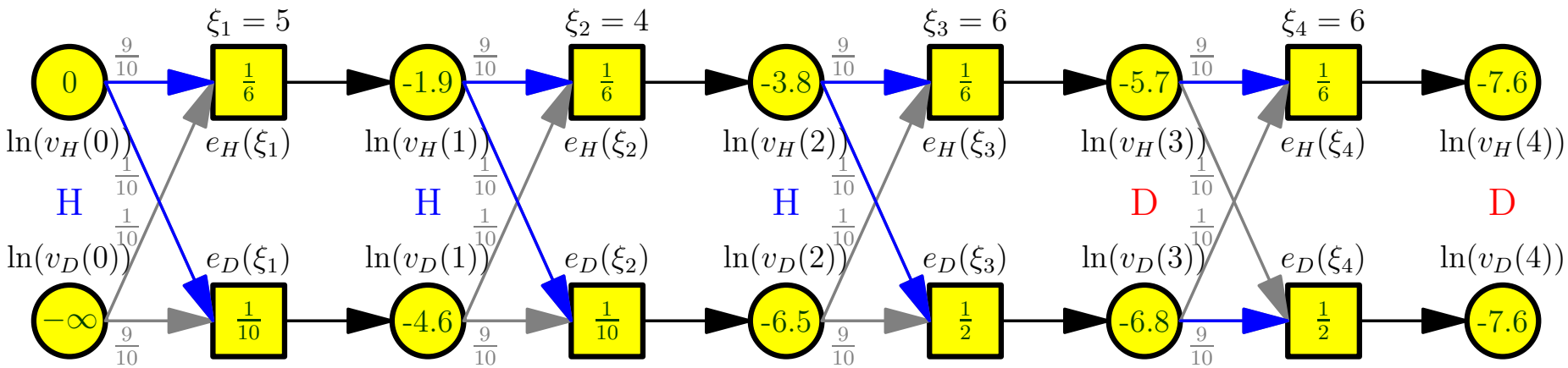

# Back to Our Dishonest Casino

- One application of the Viterbi algorithm is to make an informed guess whether the dice is fair or loaded in our dishonest casino
- Show the dice rolls, true states and Viterbi predictions

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# Probabilities of a Sequence

- The joint probability of the HMM moving through a set of states $\boldsymbol{q}$ and emitting symbols $\boldsymbol{\xi}$ is given by

$$\mathbb{P}\left(\boldsymbol{\xi}, \boldsymbol{q} | \boldsymbol{\theta}\right) = \prod_{t=1}^{T} a_{q_{t-1}\, q_t}\, e_{q_t}(\xi_t)$$

- Because the states are hidden from us we have to average them out

$$\mathbb{P}\left(\boldsymbol{\xi} | \boldsymbol{\theta}\right) = \sum_{\boldsymbol{q} \in \mathcal{Q}^T} \mathbb{P}\left(\boldsymbol{\xi}, \boldsymbol{q} | \boldsymbol{\theta}\right)$$

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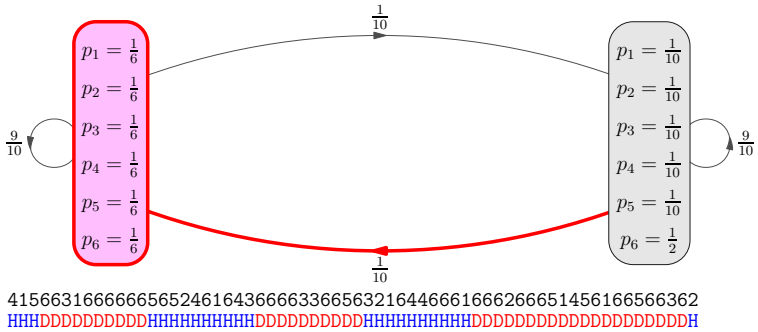
# Outline

1. Probabilistic Models
2. Hidden Markov Models
3. **Summing Over Paths**



# Summing Over Paths

- The sum of all paths involves $|\mathcal{Q}|^T$ terms which rapidly becomes intractable
- However, once again dynamic programming comes to the rescue
- Note, that previously we had used dynamic programming to find the optimal path—now we are using it to sum over all paths
- We define the forward and backward variables as

$$\alpha_t(i) = \mathbb{P}\left(\xi_1, \cdots, \xi_t, q_t = i | \boldsymbol{\theta}\right)$$
$$\beta_t(i) = \mathbb{P}\left(\xi_{t+1}, \cdots, \xi_T | q_t = i, \boldsymbol{\theta}\right)$$

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- Note, that previously we had used dynamic programming to find the optimal path—now we are using it to sum over all paths
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# The Forward Variable

- The variable $\alpha_t(i) = \mathbb{P}\left(\xi_1, \cdots, \xi_t, q_t = i | \boldsymbol{\theta}\right)$ is computed iteratively

$$\alpha_t(i) = \sum_{j \in \mathcal{Q}} \mathbb{P}\left(\xi_t, q_t = i | q_{t-1} = j, \boldsymbol{\theta}\right) \mathbb{P}\left(\xi_1, \cdots, \xi_{t-1}, q_{t-1} = j | \boldsymbol{\theta}\right)$$

$$= e_i(\xi_t) \sum_{j \in \mathcal{Q}} a_{ji}\, \alpha_{t-1}(j)$$

- With initial condition $\alpha_0(0) = 1$ (where state 0 is taken to be initial state) and $\alpha_i(0) = 0$ for $i \neq 0$

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# Using Forward and Backward Variables

- With both forward and backward variables we can compute more complex quantities

- For example, the probability of being in a particular state given a sequence $\boldsymbol{\xi}$ is

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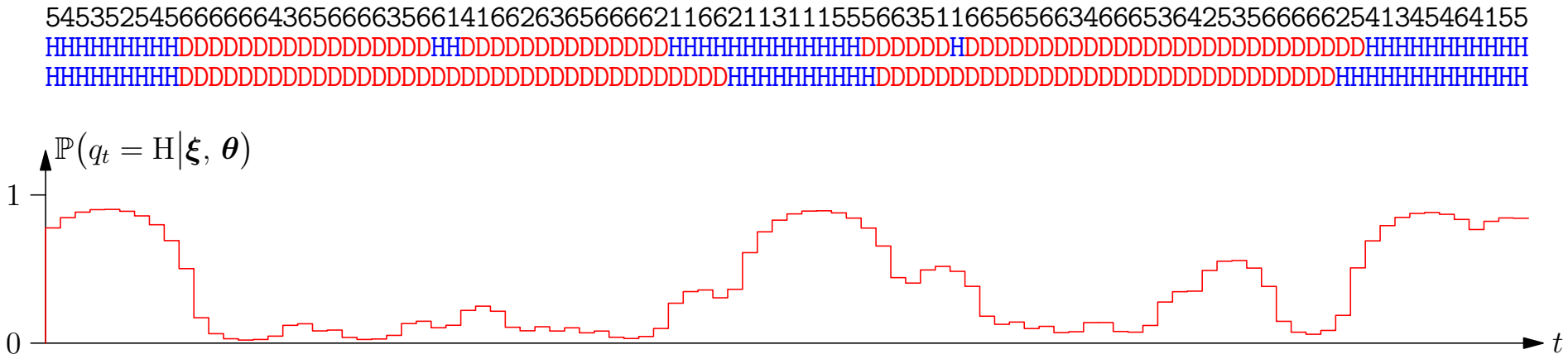
# Posterior Probability

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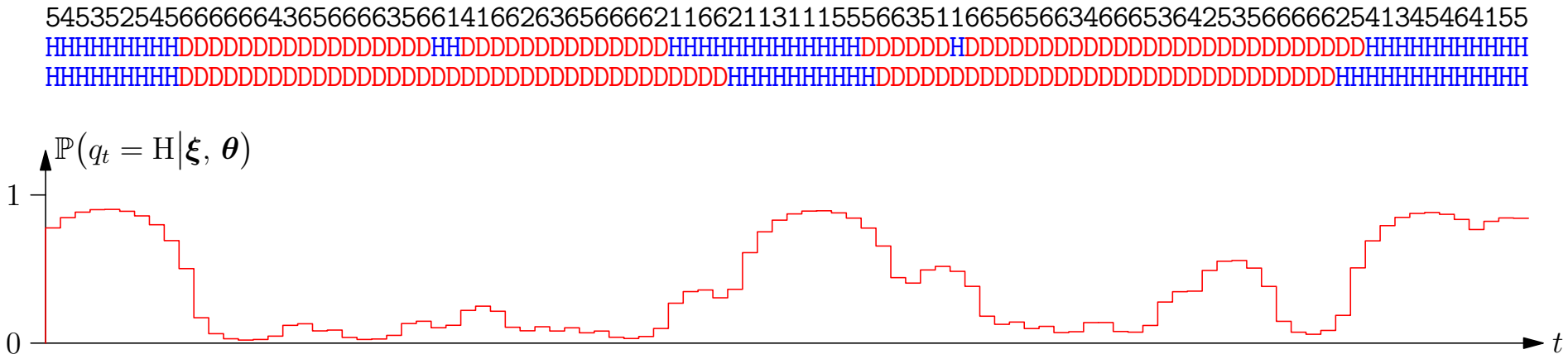


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- E.g. the most likely path as computed by Viterbi
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# Using HMMs in Biology

- In biological sequence analysis we can use an HMM, for example, to tell if a protein is a kinase protein
- If we associate different states to either an alpha helix, a beta sheet or a coil then we can calculate the probability of an amino acid being one of the above
- Of course to do this we need that our HMM has the right structure
- If we knew a lot about biology we might be able to carefully design the HMM
- In practice we very rarely do and we have to train our HMM on data—we discuss this in the next lecture

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# Summary

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- They are probabilistic Finite State Automata
- They are a generative model for sequence
- Of course, they are massively simplified generative models
- We can use dynamic programming to sum over all the unseen state

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